Time Domain Thermoreflectance

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Abstract:

Time domain thermoreflectance (TDTR) has been instrumental in measuring the heat transfer properties in bulk and nanostructured materials over the last two decades. In this Primer, we describe the optical and thermal aspects of TDTR, with an in-depth discussion on the theory, apparatus design, and implementation. We present examples that illustrate the ability of TDTR to measure thermal conductivity tensors, thermal conductance across material interfaces, and volumetric heat capacity of thin films, 2D materials and bulk materials. The ability of TDTR to spatially resolve the thermal properties of materials is useful for studying heterogenous material systems, such as materials processed in or subjected to extreme environments. We consider current limitations of pump-probe metrologies and discuss recent advancements to TDTR, such as Time Resolved Magneto-Optic Kerr Effect (TR-MOKE), beam offset TDTR/TR-MOKE, Steady State Thermoreflectance (SSTR), Frequency Domain Thermoreflectance (FDTR), and laser-flash TDTR. Finally, we present an outlook on anticipated technology development to further expand the boundaries of nanoscale thermal measurements.

[H1] Introduction

Time domain thermoreflectance (TDTR) is a laser-based experimental technique used to investigate thermal transport phenomena **[G]** across a variety of material systems [1,2]. TDTR monitors temperature-induced changes to a sample's reflectivity, known as thermoreflectance **[G]**, caused by laser heating [3–6]. In a typical experiment, the sample is coated with a thin metal film (the "transducer") that absorbs laser energy and converts it into heat. A time delayed probe beam then measures the resulting changes in reflectivity [3–6]. Finally, experimental data is compared to the predictions of a thermal model [1], with one or two key thermal property parameters adjusted until the model accurately matches the observed behavior.

TDTR confines the temperature rise to near-surface regions (spanning tens of nanometers to several micrometers [7]) by measuring the thermal response of materials on nanosecond to microsecond time-scales. This surface-localized heating allows for the precise evaluation of thermal properties of thin films, material interfaces, and micro-architectured materials (e.g., superlattices, hierarchically ordered nanocomposites, nanolaminates, etc.). These nano-systems often exhibit transport properties that differ significantly from their bulk counterparts, on account of the mean-free-paths (MFPs) **[G]** of heat carriers being altered by compositional defects, grain boundaries, and strain, etc. [8–14]; TDTR has been critical in identifying these effects.

The high lateral resolution of TDTR, typically ranging from 1 – 20 μm, further enhances TDTR's versatility. This fine resolution enables the detailed mapping of microstructural inhomogeneities or composition gradients[15–21], and facilitates the characterization of thermal anisotropy in bulk and thin-film systems, including 2D materials [22–25], multilayers [26–32] , and porous organic frameworks [25,33–35]. Moreover, TDTR's lateral resolution makes it possible to measure samples with sub-millimeter dimensions, e.g., exfoliated flakes of 2D materials [22–25], small crystals of next-generation semiconductors [36–41], and single polymer fibers [42]. Combined with its non-contact nature and relatively simple sample preparation requirements, TDTR is a powerful tool for assessing the thermal properties of a wide range of materials and devices with complex geometries[43–53]. In short, TDTR and its offshoots have emerged over the past two decades as a robust, nearly universal method for measuring the thermal properties of almost any material that can be prepared with an optically smooth surface.

[H2] Evolutionary Timeline of TDTR:

Early innovations in pump-probe thermoreflectance have laid the groundwork for today's nearly standardized TDTR system, where both time- and frequency-domain thermal responses are exploited to enable thermal property measurements. Eesley [54,55], Paddock and Eesley [56], Opsal *et al*. [57], Opsal and Rosencwaig [58], Thomsen *et al*. [59], Stoner and Maris [60] and Bonello *et al.* [61] were amongst the first experimentalists to use pump-probe thermoreflectance techniques to study thermal and mechanical properties of thin films and interfaces. Early studies by Capinski and Maris [62–64] and Bonello *et al.* [61] recognized that, with the implementation of high repetition rate lasers (such as the now-standardized ~80 MHz Ti:sapphire oscillators, featuring < 200 fs pulse widths), it becomes necessary to account for the effects of pulse accumulation **[G]** on the measured signals. The temporal spacing between pulses, $1/f_{\mathrm{rep}}$, where f_{rep} is the laser repetition rate, is much shorter than the time that the sample needs to cool to equilibrium after heating. As a result, heat from each pulse accumulates. While these early studies recognized that pulse accumulation occurred, they still focused their analyses exclusively on the time-domain (or the "in-phase") response to pulsed heating. Since heat diffuses only a few tens to hundreds of nanometers over nanosecond time-scales, this time-domain approach limits the sensitivity of measurements to the thermal properties of the underlying sample.

In the early 2000s, Cahill and co-workers introduced a breakthrough by decoupling the time-domain response from the accumulated signal [2,65–67]. Cahill mathematically demonstrated that an additional "out-of-phase" component, which relates the sample's thermoreflectance response to the modulated heating from the pump, could be analyzed alongside the traditional in-phase response[1]. This breakthrough enabled pump-probe thermoreflectance measurements to access a wider range of time and length scales. Today, this innovation underpins the nearly-standardized configuration of modern TDTR experiments, and it is an important and powerful distinction between TDTR and other transient pump-probe techniques. Inspired by the definitions put forth by Koh *et al.* [68], we suggest that the term "TDTR" be reserved for experimental techniques where these in-phase and out-of-phase responses are used, while pump-probe techniques based on low-repetition rate lasers or continuous wave (CW) probe lasers be termed "transient thermoreflectance" (TTR).

[H2] Primer Outline:

The primary focus of this Primer is to provide details of the experimental and analytical elements that define modern-day TDTR experiments. We first discuss the experimental setup for TDTR and its variants. Then, we review principles for common measurements, such as thermal interface conductance, thermal conductivity tensors, and heat capacities. A secondary focus of this Primer is to discuss advanced TDTR methods and related metrologies designed to address specific measurement challenges. For example, to measure the three-dimensional (3D) thermal conductivity tensor of anisotropic materials, beam-offset TDTR uses offset pump and probe laser beams to resolve lateral heat spreading [69], see Fig. 1(b). Time-resolved magneto-optic Kerr effect (TR-MOKE) leverages magneto-optic thermometry to improve sensitivity to in-plane and cross-plane k of thin-films [70–72], see Fig. 1(c). Laser-flash TDTR allows measurements of the spatial evolution of the temperature profile in the through-plane direction as a function of time and heating frequency [73,74], see Fig. 1(d). A comparative table outlining the operating principles, advantages, and measurement limitations of TDTR and all of its variants is provided in Supplementary Table 1, along with critical sample preparation requirements for each. Next, we examine studies showcasing TDTR applications across diverse material systems. We provide best-practices for data reproducibility, identify sources of error, and consider experimental design factors. The Primer then explores the inherent limitations of TDTR in sampling representative volumes of material beneath a surface, and compare to alternatives such as frequency-domain and steady-state thermoreflectance (FDTR and SSTR, respectively). Finally, the Outlook section explores emerging trends, and considers future directions for TDTR and nanoscale thermal metrology.

To make the primer accessible to a broad audience, a number of basic TDTR principles are reviewed here that were covered in the excellent TDTR tutorial written by Jiang *et al*. [75]. In addition, this primer also covers a number of other topics, including an explanation of the physical meaning of TDTR signals, laser-flash TDTR, and SSTR. The primer also reviews different TDTR applications than Jiang *et al.* [75], e.g. ion irradiation, liquids and biomaterials, etc.

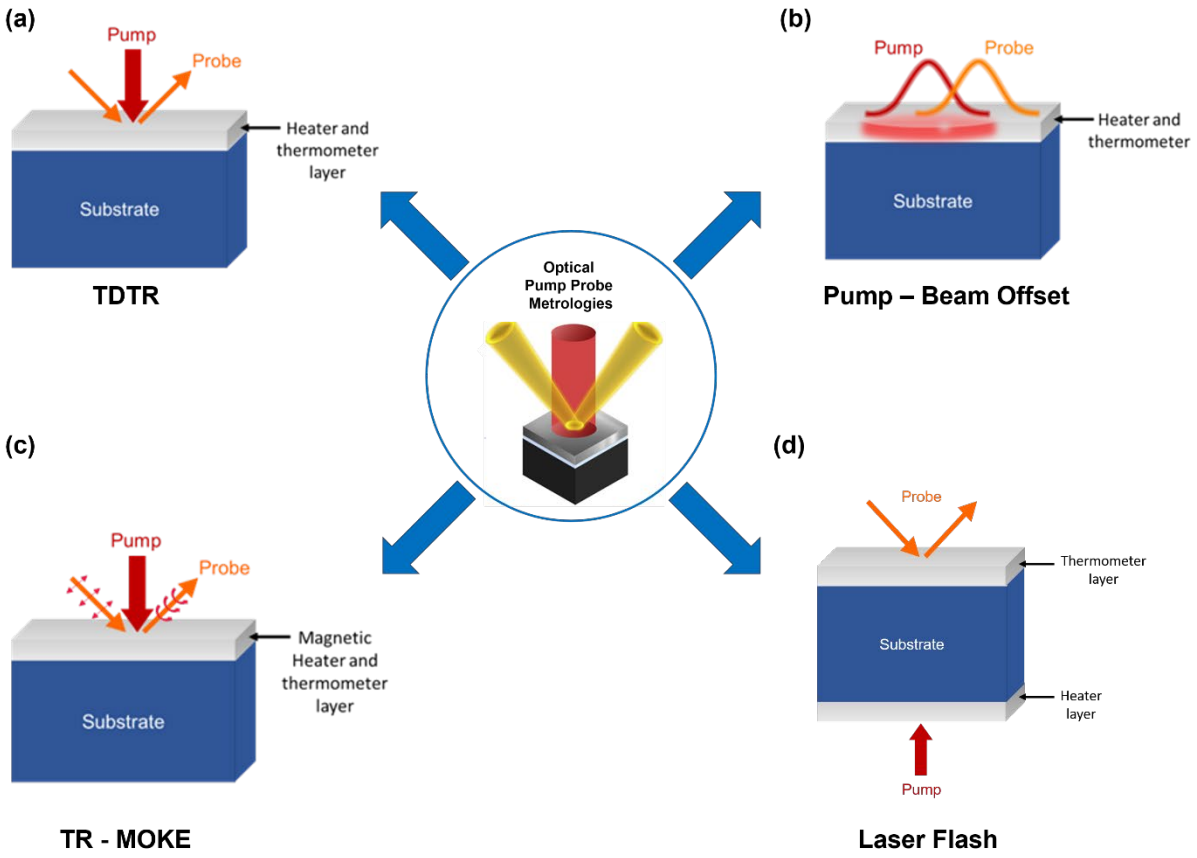


Figure 1: Introduction to Optical Pump-Probe Metrologies for Thermal Property Measurements. (a) Time Domain Thermoreflectance (TDTR) measurements. This geometry allows thermal conductivity and heat-capacity measurements of thin-films and bulk samples. (b) Beam offset measurements, where the heater and thermometer spots are offset by a specified distance. This geometry allows measurements of thermally anisotropic samples. Beam offset measurements can be performed in both, TDTR and TR-MOKE configurations. (c) Time-Resolved Magneto-Optic Kerr Effect (TR-MOKE) measurements. By substituting thermoreflectance with magneto-optic thermometry, this technique improves spatiotemporal resolution and allows measurements of thinner films. (d) Laser-flash measurements, where the heater and thermometer spots are on opposite sides of the substrate. By providing additional information about the temperature-profile in the through-plane direction, laser flash is useful for multilayer samples with multiple unknown properties.

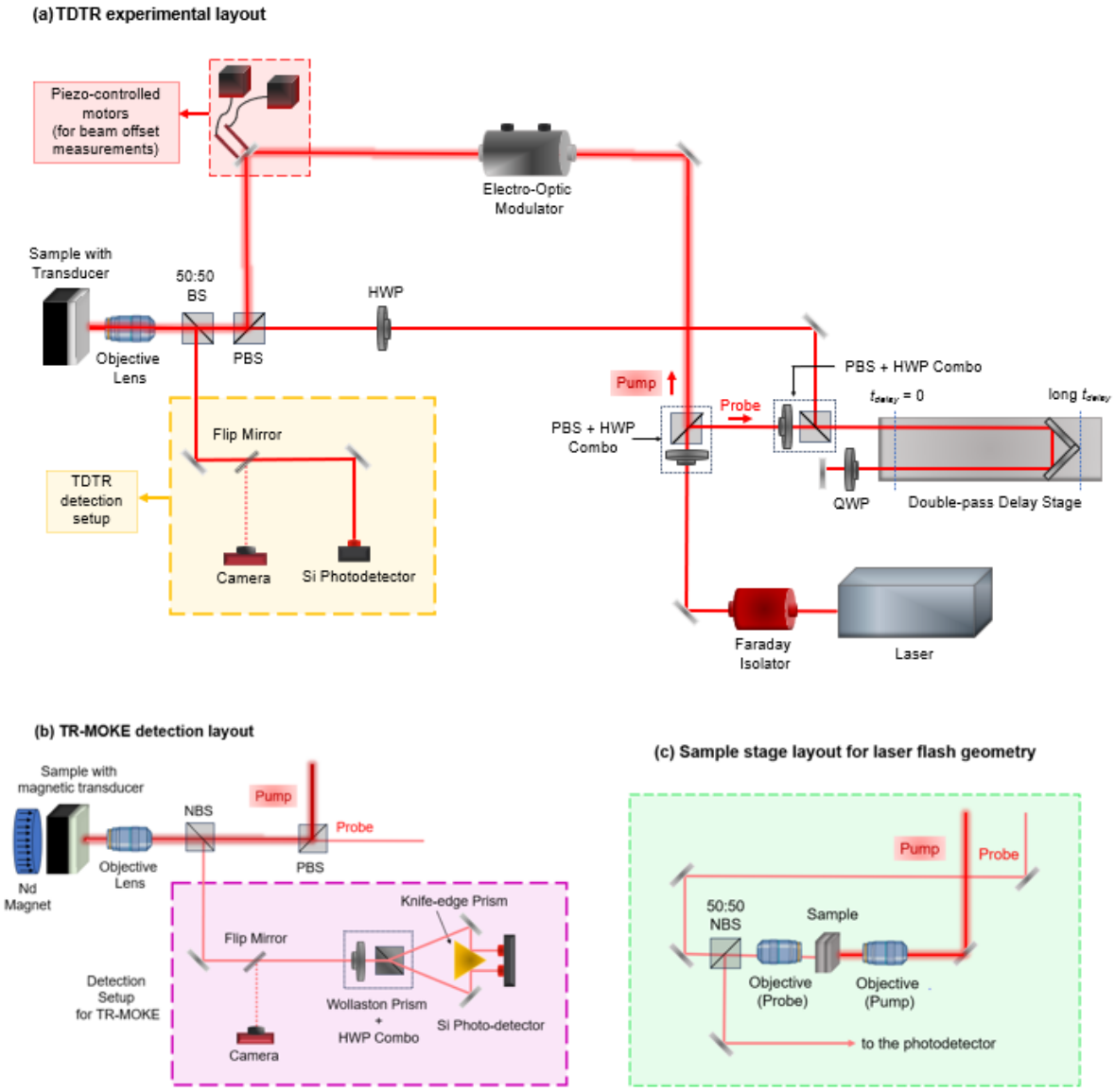


Figure 2: Schematic for Optical Apparatus enabling TDTR and Variant Metrologies. (a) TDTR experimental layout. [Abbreviations in the Figure: QWP (quarter wave-plate), HWP (Half wave-plate), PBS (polarized beam-splitter), BS (beam-splitter), NBS (normal beam-splitter)]. A more detailed TDTR layout is shown in Fig. S1 in the Supplementary Material. (b) The detector layout for TR-MOKE experiments, and (c) the sample stage design layout for laser-flash TDTR experiments.

[H1] Experimentation

Here, we provide an overview of the apparatus required for TDTR, and describe the basic theory underpinning TDTR signals. Figure 2 shows the optical layout for a TDTR experiment, along with proposed setup modifications that enable multiple measurement configurations in a single optical table. We provide a detailed step-by-step guide for building TDTR setups in Supplementary Note 1.

[H2] TDTR Apparatus:

[H3] Laser System. TDTR experiments typically employ a solid-state diode-pumped Ti:sapphire oscillator that generates sub-picosecond pulses at a central wavelength of ~800 nm, with a spectral bandwidth of ~10 nm and a repetition rate of 80 MHz. If the output beam is elliptical, cylindrical lenses can be used to correct it [69]. In Supplementary Box 1, we provide a primer on Gaussian beam optics and associated formulae, that are useful for beam shaping, beam collimation, and focusing down to diffraction-limited spot-sizes.

[H3] Delay Stage. The delay-stage controls the time-delay between the pump and probe beams at the sample, by introducing a controllable difference in path length. The delay stage can be placed either in the pump path or the probe path. Setting the delay stage on the pump path introduces a phase shift, $e^{i.2\pi.f_{mod}.t_{delay}}$, in the heating function that needs to be accounted for in the thermal modelling, where f_{mod} is the pump modulation frequency [69]. The delay time is equal to the optical path length difference, divided by the speed of light. The delay stage needs to be able to introduce a path length difference of at least ≈ 0.9 m, yielding a time-delay of at least ≈ 3 ns, so that the transducer/sample interface conductance, G, can be determined. G is measured by observing the time-scale over which the metal film transducer's temperature decays, $\tau = h_t C_t/G$. For typical values, e.g. $h_t = 10^{-7}$ m, $C_t = 3 \times 10^6$ J m^{-3} K^{-1}, and $G = 10^8$ W m^{-2} K^{-1}, τ is ≈3 ns.

[H3] Modulation of the Pump Beam. Modulation of the pump beam is necessary, because TDTR derives its sensitivity to the thermal properties of the sample by measuring the thermal response of the sample to harmonic heating at f_{mod}. Typical modulation frequencies range from 0.1 to 20 MHz. An electro-optic modulator (EOM) is used to modulate the train of pump pulses. The resulting heating function is shown in Fig. 3(c). In Supplementary Note 1, we discuss the effect of square vs. harmonic modulation envelopes.

[H3] Objective Lens and Integrated Microscope. The pump and probe beams are focused onto the sample surface using an objective lens. Lenses with long working distances facilitate integration with sample environments such as optical cryostats or heater-stages for *in situ* measurements. The probe beam must not be clipped at the back focal plane of the objective lens, or anywhere in the optical path between the sample and detector. Clipping the beam causes measurement artifacts by converting thermal-expansion induced phase modulations of the probe beam into intensity modulations[76–81]. This condition limits the size of the probe beam in free-space, and consequently, the diffraction-limited spot-size on the sample – please refer to Supplementary Box 1 for associated formulae. An "integrated microscope", composed of a CCD camera and a suitable lens, allows for sample surface imaging and ensures that the sample is reliably placed at the focal plane of the objective lens. Imaging of the surface also allows users to avoid regions of the sample with dust or defects, and simplifies the spatial overlap of the pump and probe beams on the sample surface[82].

[H3] Detector and Lock-in Amplifier. The reflected probe beam is focused onto a Si photodetector with a response time sufficient to resolve signals at f_{mod}. The detector is connected to a radio frequency (RF) lock-in amplifier. The lock-in amplifier measures the amplitude and phase of the AC signal at f_{mod}. In Supplementary Note 1, we discuss design considerations for the detection circuit, and methods for blocking the pump beam from reaching the detector.

[H2] Advanced TDTR Methods:

[H3] Beam Offset Measurements. Precise control over the spatial overlap of pump and probe beams, i.e., the beam offset, is achieved by using either a goniometer with motorized actuators on the PBS (polarized beam-splitter) in front of the objective lens [69], or by utilizing motorized actuators on a mirror in the pump beam path (as shown in the red transparent box in Fig. 2(a))[83]. Both of these methods deflect the pump beam by an angle of $\Delta\theta$ as it enters the objective lens, and this produces a resulting beam offset of $\Delta x_0 = F\Delta\theta$ on the sample surface, where F is the focal length of the objective lens. We discuss beam offset TDTR measurements in more detail in the Results section.

[H3] Time-Resolved Magneto-Optic Kerr Effect (TR-MOKE). The primary distinction between the TDTR and TR-MOKE experimental setups lies in the detector optics, as illustrated in Fig. 2(b). TR-MOKE measures the magneto-optic Kerr effect, which describes the change in

polarization of the reflected probe beam caused by the magnetization of a ferromagnetic transducer. To measure the change in polarization, the reflected probe is split into orthogonally polarized components with a Wollaston prism and directed onto a balanced photodetector. Prior to the measurement, a half wave plate before the prism is rotated to balance the detectors. Then, during the measurement, any temperature-induced changes in polarization imbalances the detectors, and generates a signal proportional to the temperature change. To suppress any thermoreflectance signals, it is useful to repeat the measurement with the transducer oppositely magnetized and taking the difference of the two signals. The resulting TR-MOKE ratio of in-phase to out-of-phase voltages can then be analyzed similarly to a conventional TDTR measurement.

As illustrated in Fig. 2(b), TR-MOKE measures the real part of the ferromagnetic metal's Kerr rotation angle. Signal can sometimes be improved by instead measuring the imaginary part of the Kerr angle, or some linear combination of the two. This can be accomplished by replacing the half wave plate before the prism with a quarter wave plate [84], or by adding a Babinet-Soleil compensator before the half wave plate [83]. We discuss TR-MOKE in more detail in the Results section.

[H3] Laser Flash TDTR. Laser-flash TDTR involves heating and probing a sample on opposite sides to measure the cross-plane heat-transfer as a function of time, and/or heating frequency. Thus, the optical layout needs to be modified as shown in Fig. 2(c). The pump beam path remains identical to that in a standard TDTR experiment, whereas the probe beam is redirected using mirrors to reach the opposite side of the sample. The resulting reflected probe is then directed to a photodetector. We discuss laser-flash TDTR in more detail in the Results section.

[H2] Signal Acquisition and Processing:

The voltage signal measured by the RF lock-in amplifier is given by:

$$V(t) = \varsigma\, P_{probe}\, \frac{dR}{dT}\, \Delta T(t) \quad (2.1)$$

Here, ς represents a setup-specific constant that depends on several factors, including the optical transmissivity of all the optical elements between the sample and the detector, the detector's responsivity, and the gain of the electronics. P_{probe} is power of the probe beam, while $\frac{dR}{dT}$ is the thermoreflectance coefficient of the transducer.

Lock-in signals (i.e., the in-phase voltage signal, V_{in}, and the out-of-phase voltage signal, V_{out}) are collected as a function of pump-probe delay time. Initially, data is collected over a short time-delay window, e.g., -20 to 100 ps, in small increments (~1 ps). This initial linear scan allows for the lock-in phase offset to be set, and provides picosecond acoustics **[G]** signals for film thickness determination [85]. (Picosecond acoustic signals require pulse durations on the order of 1 ps or less.) Subsequent data is collected on a log-scale from ~100 ps to the maximum delay time (e.g. ~ 4 ns). See Supplementary Fig. S2 for an outline for a basic TDTR measurement.

The lock-in measures the phase relative to a reference signal at f_{mod}, which is provided by the function generator. TDTR analyses requires us to know the phase of the signal relative to the pump heating function. Since the phase of the pump heating function and reference signal will be different, a phase correction must be applied. This can be done during data analysis, or by adjusting the lock-in's phase setting. The optimal phase offset will flatten the out-of-phase signal near zero delay time, i.e., make V_{out} constant from -20 to 20 ps. This is because the out-of-phase signal describes the thermal response on $1/f_{\mathrm{mod}}$ time-scales (explained in the next section), which are not affected by a delay-time change of 40 ps. We discuss rules of thumb for choosing a suitable modulation frequency in Supplementary Note 2.

[H2] Signal Theory:

We now turn our attention to understanding and modelling the measured TDTR signals. The ratio of the V_{in} and V_{out} signals is analyzed by comparing it with the predictions of the multilayer thermal model **[G]** described by Cahill[1]. Prior studies provide detailed descriptions of the math underpinning the multilayer thermal model for TDTR data analysis [1,7,70,86–88]. Therefore, we limit our focus to building a physical intuition for the V_{in} and V_{out} signals.

TDTR's effectiveness derives from its ability to measure the response of the system to both pulsed heating and harmonic heating. We illustrate how this works in Fig 3 by comparing thermal responses of a semi-infinite solid with thermal effusivity of $\approx 5 \times 10^3$ W s$^{0.5}$ m^{-2} K$^{0.5}$ from heating by a single laser pulse (Fig. 3(a)), harmonic heating (Fig. 3(b)), and the heating function in a TDTR experiment (Fig. 3(c)). In Supplementary Note 2, we provide further details about the relationship of these three types of heating.

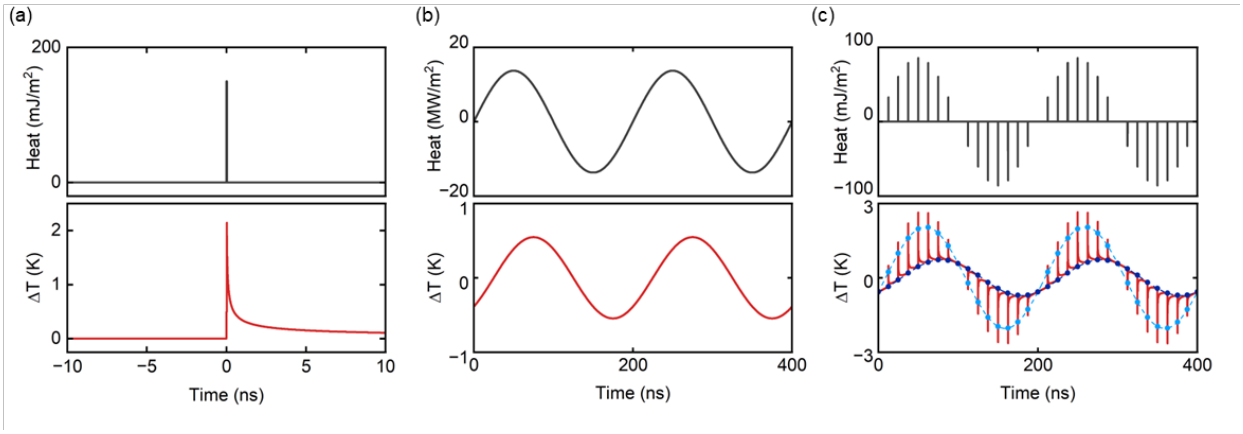


Figure 3: Thermal responses to different types of heating. (a) Transient temperature rise following a single pump pulse. (b) Temperature oscillation cause by harmonic heating at 5 MHz; the temperature waveform (red) lags the heating function. (c) Temperature response to a pulse train whose envelope is modulated at 5 MHz (red curve). In a TDTR experiment, measurements are collected over a series of delay times to gain information about both the harmonic response and pulsed response. The probe samples the instantaneous temperature at fixed delays after each pump pulse: dark blue markers at -100 ps and light blue markers at +100 ps. The -100 ps trace is similar to the harmonic temperature response in (b). It describes the thermal response of the sample on a $1/f$ time-scale. Alternatively, the amplitude of the waveform at +100 ps reflects the pulsed response shown in (a) at a time 100 ps after the pulse.

The V_{in} signal in a TDTR experiment corresponds to the temperature response synchronized (in-phase) with the pump modulation. It reflects the immediate temperature rise and the energy stored in the sample after each pump pulse. In other words, V_{in} describes the thermal response of the sample to heating from a single pulse, as illustrated in Fig. 3(a). When the metal film (transducer) absorbs a single laser pulse, the magnitude of this temperature rise depends on the transducer's thickness and heat-capacity. Subsequently, the transducer cools as heat flows into the sample beneath. The characteristic cooling time, τ, is governed by the interfacial thermal conductance G and the sample's thermal conductivity k. Thus, the in-phase signal decays with time, and has an amplitude that is inversely proportional to the transducer's heat capacity per unit area: $V_{in} \propto f(t)/h_t C_t$, where the function $f(t)$ describes the decay rate. For samples with $k >> h_t G$, the transducer can be treated as a lumped capacitance, and the temperature decays exponentially: $f(t) = \exp(-t/\tau)$.

Alternatively, the V_{out} signal lags the pump by 90°, and thus, captures the delayed thermal response governed by heat diffusion away from the heated region. The sample acts as a thermal integrator. Energy is deposited in discrete pulses but diffuses away slowly. As a result, V_{out} closely resembles

what would be observed in response to continuous harmonic heating at frequency f_{mod} (Fig. 3(b) and 3(c)). In response to harmonic heating, the sample's temperature oscillates with an amplitude proportional to the thermal resistance of the sample across a characteristic length *d*, defined as the distance that the heat diffuses during one modulation period $1/f_{mod}$. For bulk samples, this distance is the thermal penetration depth $d_p = \sqrt{k/\pi C f_{mod}}$ and $V_{out} \propto d_p/k$. Or, equivalently, V_{out} is inversely proportional to the sample's thermal effusivity: $V_{out} \propto 1/\sqrt{kC}$ (in Supplementary Note 2, we provide a more rigorous derivation of this heuristic).

In TDTR experiments, the ratio of V_{in} and V_{out} is analyzed, which is convenient because it is independent of a number of experimental parameters that both V_{in} and V_{out} are proportional to, including ς, the transducer's dR/dT, and the pump and probe powers. Based on the discussion above, the ratio for a TDTR measurement of a high k bulk sample can be approximated as

$$-\frac{V_{in}}{V_{out}} \approx \frac{2\sqrt{\pi f_{mod}}}{f_{rep}} \frac{\sqrt{kC}}{hC_t} \exp(-t/\tau). \quad (2.2)$$

For low k samples, the amplitude will be the same, but $\exp(-t/\tau)$ will be replaced by a different decay function, $f(t)$, that depends on $h_t C_t, G$ and $\sqrt{kC}$. A more detailed discussion and derivation of Eq. (2.2) is provided in Supplementary Note 2. The key take-aways from Eq. 2.2 are that (i) the ratio signal in TDTR experiments usually scales with the sample's effusivity, and (ii) the decay rate of the ratio signal provides a sensitive measure of the interface conductance between the transducer and interface. We emphasize that, while Eq. 2.2 is a useful heuristic for considering how changes in thermal properties will alter TDTR signals, actual data analyses is performed with the multilayer thermal model described in Ref. [1].

Frequency domain thermoreflectance (FDTR) is a closely related technique to TDTR. Both measure the response of the sample to harmonic heating. In FDTR, the phase-lag between the heating function and thermal response (see Fig. 3(b)) is measured as a function of heating frequency. In TDTR, the response to pulsed heating is measured as a function of time-delay, normalized by the out-of-phase thermal response to harmonic heating at f_{mod}.

[H2] Sample Preparation / Choosing a Transducer:

TDTR experiments usually require coating the sample with a metal transducer. The sample surface before and after transducer deposition needs to be smooth (< 10 nm). A metal transducer (> 50 nm

thick) in TDTR experiments confines heating and temperature sensing to the surface, prevents electron-hole pair excitation, and simplifies thermal modeling by eliminating the need for depth-dependent absorption and thermoreflectance calculations [73]. Supplementary Note 3 provides details about the role of the transducer, and factors to consider during selection.

The figure of merit for maximizing signal-to-noise is $\frac{1}{R} \times (dR/dT)$ of the metal at the probe wavelength [89,90]. This value typically peaks at wavelengths near interband optical transitions [91]. Wang *et al.* [89] and Wilson *et al.* [90] report dR/dT for a variety of transducers. Aluminum is a good transducer for most experiments with a probe wavelength near 800 nm, where it has large thermoreflectance, strong picosecond acoustic signals, and high k_t . It also has good adhesion to most samples.

For in-plane thermal conductivity (k_r) measurements, the transducer k_t should be low. A high k_t transducer shunts the in-plane heat current. This reduces sensitivity to the sample's in-plane conductivity. Specifically, the transducer's in-plane conductance ($h \cdot k$) should be lower than the sample's in-plane conductance. The in-plane conductance of a thin-film is $h_{film} \cdot k_{film}$, while for bulk samples it is $d_p \cdot k_{substrate}$. A metal alloys like NbV, ($k \sim 20$ W m^{-1}K^{-1}) [70] or a AlTi bilayer ($k \sim 40$ W m^{-1} K^{-1}) [92] combine low k with high dR/dT, which make them good choices for in-plane measurements. The opportunity exists to conduct TDTR measurements without metal film transducers under specific experimental and material conditions, which we discuss in Supplementary Note 3.

[H1] Results

In this section, we describe measurement guidelines and results for the common types of TDTR experiments. These include measurements of thermal interface conductance, cross-plane thermal conductivity k_z, in-plane thermal conductivity k_r, and the volumetric heat capacity C.

[H2] Thermal Model Parameters:

Supplementary Fig. S2 shows the standard TDTR experimental procedure. Analyzing data with the thermal model requires layer-specific parameters: h, C, and k_z. If d_p and w_0 are comparable, heat transport becomes 3D (see Fig. 4(d)), and k_r is also required. This case is discussed later. The interfacial thermal conductance (G) between layers is also needed. Finally, the laser spot-size (w_0) must be known. Typically, a single TDTR experiment measures two quantities: the ratio of pulsed to harmonic heating responses, $-V_{in}/V_{out}$, and its decay rate with time-delay. Since only two experimental quantities are measured, *in most cases*, it is only possible to fit for two parameters when analyzing a TDTR data set at a single modulation frequency: the transducer/sample G, and one additional parameter, such as the sample's k. All other parameters in the model must be independently determined. The parameters being fit should not be strongly coupled. For example, looking at Eq. 2.2, we expect k and C to be coupled in standard 1D TDTR measurements, because they both effect the ratio in the same manner. Supplemental Fig. S4 shows the results of calculations done with the TDTR thermal model to determine whether two parameters are coupled.

For a 1D TDTR measurement ($w_0 >> d_p$, Fig. 4(c)), the ratio is most sensitive to the thermal capacitance of the transducer (denoted by $h_t C_t$) [93]. For TDTR measurements with a 3D temperature-profile ($w_0 \sim d_p$, Fig. 4(d)), the ratio is most sensitive to the laser spot-size (w_0) [69,93]. Thus, these parameters must be precisely known or measured, so as to minimize uncertainties in the reported k and G values.

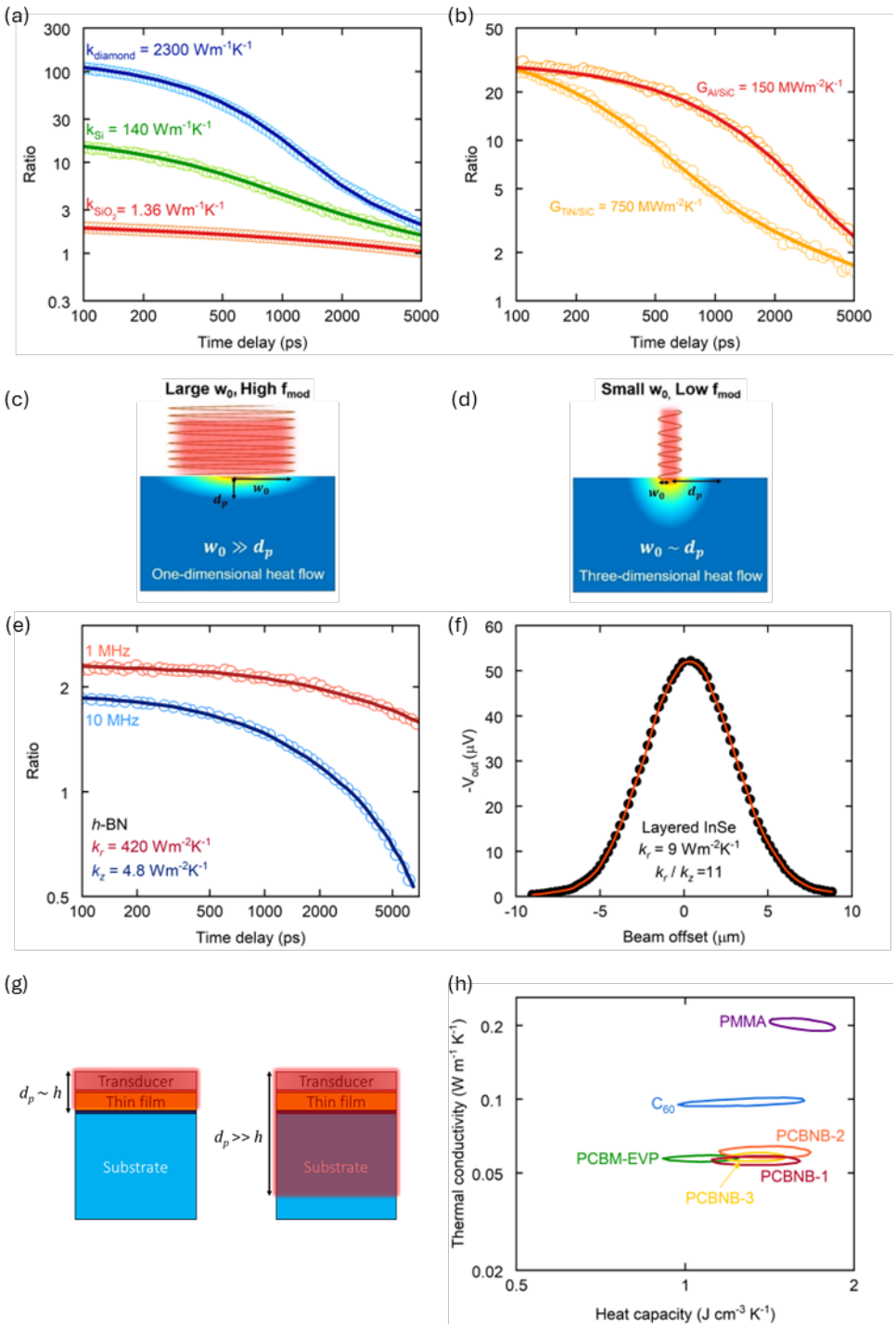


Figure 4: Representative data for most common TDTR experiments. (a) Effect of sample thermal conductivity on TDTR signals. Markers are TDTR data for TiN coated diamond, Si, and SiO_2 [94]. The lines are model predictions with *k* and *G* treated as fit parameters. The magnitude of the ratio depends on the substrate's thermal conductivity, increasing with higher conductivity. (b) Effect of the interface conductance on TDTR signals. The open markers are TDTR data for Al/4H-SiC and TiN/4H-SiC. The decay of the ratio varies with the thermal interface conductance between the transducer and substrate, with higher conductance corresponding to a faster decay [94]. (c) and (d) Impact of modulation frequency and spot size on heat flow regimes in TDTR experiments. In (c), at high modulation frequencies and large

spot sizes, TDTR measures thermal effusivity under one-dimensional heat flow conditions. Conversely in (d), at low modulation frequencies and small spot sizes, TDTR captures thermal diffusivity in a three-dimensional heat flow regime. (e) TDTR data and thermal model fits for h-BN measured with a 10 μm spot size at modulation frequencies of 1 MHz and 10 MHz. The 10 MHz data was fit to extract the through-plane thermal conductivity, while the 1 MHz data was fit for the in-plane thermal conductivity [95]. (f) Normalized out-of-phase signal as a function of beam offset for a 297 nm thick InSe flake (circles), with Gaussian thermal model fit (solid red line) [96]. (g) Impact of modulation frequency on the thermal penetration depth relevant to heat capacity measurements of thin films. When $d_p < h$, the TDTR ratio is sensitive to both the heat capacity and thermal conductivity of the thin film. When $d_p >> h$, the heat capacity of the thin film become irrelevant, and the ratio is sensitive to the thermal conductivity of the film. (h) Contour plots showing combinations of thermal conductivity and heat capacity that yield good fits by analyzing measurements at both low and high modulation frequencies for C_{60}, PMMA, PCBM, highly disordered PCBNB-1, crystalline phase I PCBNB-2, and crystalline phase II PCBNB-3 [97].

[H2] Measurements of Thermal Interface Conductance and Cross-plane Thermal Conductivity:

The effect that a sample's k has on TDTR signals is shown in Fig. 4(a) by comparing data and model fits for bulk diamond, silicon and amorphous SiO_2 with TiN transducers. The differences in k for these samples translates into ratios that vary by about a factor of 40, because $-V_{in}/V_{out}$ scales with $\sqrt{k}$, (see Eq. (2.2)). In addition to providing k, the decay of the ratio with time-delay provides a sensitive measure of G = 500 and 400 MW m^{-2} K^{-1} for TiN/Si and TiN/Diamond, respectively[94] (note, for SiO_2, the low thermal effusivity of the substrate prevents the metal film from cooling appreciably, and it is not possible to accurately fit for G in addition to k).

To show the effect that G has on the ratio, in Fig. 4(b) we compare the results of TDTR measurements of 4H-SiC with two different metal transducers: Al and TiN. Since both curves are TDTR measurements of 4H-SiC, the amplitudes of the ratio in both measurements are the same at short delay times. However, the thermal interface conductance differs by a factor of 5, so for the TiN/4H-SiC measurement the ratio decays much more rapidly with delay time t.

[H2] Thermal Conductivity of Thin Films:

TDTR is effective at characterizing the cross-plane transport properties of thin-films. In general, TDTR measurements of thin-films involve the determination of three thermal properties: the transducer/film interface conductance G_1, the film thermal conductivity k_f, and the film/substrate interface conductance G_2. The ability of a TDTR measurement to independently determine these parameters depends on how h_f (film thickness) compares to two other key length scales: d_p and

h_t (transducer thickness). We consider three limits, and describe heuristics that are useful for designing experiments.

1) When h_f is less than h_t, TDTR measures a single lumped resistance: $1/G_1 + h_f/k_f + 1/G_2$. The strong coupling of the three parameters of interest occurs because TDTR works, in part, by analyzing the time-scale for cooling of the transducer after pulsed heating. For the transducer to cool, it must transfer heat into an adjacent region of comparable thickness [98]. Therefore, TDTR's depth resolution is inherently limited by h_t, making it challenging to resolve thermal resistances localized to regions thinner than h_t.

2) When h_f is greater than h_t, but less than d_p, the sensitivity of TDTR to k_f is coupled only to G_2. In this scenario, TDTR also measures an effective conductivity (k_{eff}): $h_f/k_{eff} = 1/G_2 + h_f/k_f$. The reason it is possible to separately determine k_{eff} from G_1 is because the sensitivity to G_1 is derived from the decay rate of ratio with delay time (see τ in Eq. 2.2). Alternatively, k_{eff} is being deduced the amplitude of the ratio signal (see k in Eq. 2.2). The differing depth-sensitivities of V_{in} and V_{out} make it possible to independently determine G_1 and k_{eff}.

If the film's resistance h_f/k_f is much larger than the thermal interface resistances, it does not matter that these parameters are coupled, and $k_{eff} \approx k_f$. For k_{eff} to be within 20% of k_f, it is necessary for the film resistance to be at least ~5x larger than the thermal interface resistances, which occurs when $h_f > 5k_f/G_2$. For a typical interface conductance value of G_2 ≈200 MW m^{-2} K^{-1}, the minimum thickness for a film with k_f of 1, 3, and 10 W m^{-1} K^{-1} is 25, 75, and 250 nm.

3) If $h_f > d_p$, the heated region is confined to the film, so G_2 and the substrate conductivity (k_{sub}) have minimal impact on the observed signals. The maximum frequency possible in a TDTR experiment is typically 20 MHz [99]. Assuming a typical value for the film heat-capacity of 2 MJ m^{-3} K^{-1}), the minimum film thickness for films with k_f of 30, 100, or 300 W m^{-1} K^{-1} is 500, 900, and 1500 nm. (When measuring a high k_f on a low k substrate, the minimum film thickness increases, e.g. to $1.5d_p$ for Cu on SiO2 [98].)

In the above discussion, we have described two minimum thickness criteria: the film should be thick enough that its thermal resistance dominates, or the film should be thicker than the thermal penetration depth. Setting the two minimum film thickness criteria equal to one another yields

$k_f = G_2^2/(25\pi C_f f_{mod})$. Using the values described above for G_2, C_f, and f_{mod} yields $k_f \approx 13$ W m^{-1} K^{-1}. For films with k_f less than this value, it is easier to satisfy dominant resistance condition. For films with k_f larger than this value, it is easier to satisfy the penetration depth condition.

For high k_f films with $h_f > d_p$, it is sometimes possible to determine G_2 with a second TDTR measurement with a lower f_{mod} [100]. Then, the high frequency measurement is analyzed to determine k_f and G_1, and the lower frequency measurement can be analyzed with only G_2 as a fit parameter. However, this approach requires the interface resistance $1/G_2$ to be comparable to $1/G_1$ and h_f/k_f, otherwise the uncertainty in G_2 will be large.

For scenarios where TDTR measures an effective film conductivity, the value of k_f can sometimes be isolated by measuring samples of various thicknesses [74,101–103]. The slope of total resistance versus thickness provides the intrinsic film conductivity, separating bulk from interface effects. However, this approach assumes the films of varied thickness have identical k_f, which is not always true. Crystal quality, and therefore k_f, can increase with film thickness because grains coalesce and defect densities decrease.

[H2] Frequency-Dependent In-Plane Thermal Conductivity Measurements:

TDTR measurements become sensitive to k_r when d_p is comparable to w_0, because the in-plane temperature gradients become comparable to the through-plane gradients, see Fig. 4(d). This allows measurements of k by performing at least two TDTR measurements at different f_{mod} [100]. An example of this type of measurement is shown for h-BN in Fig. 4(e) [95]. A standard TDTR measurement with $w_0 >> d_p$ provides a measurement of k_z and G. Then, a second measurement where w_0 is comparable to d_p allows the value for k_r to be determined. Sensitivity to k_r is maximized when $d_p = w_0$ [104]. We note that for TDTR measurements where w_0 is comparable to d_p, the predictions of the thermal model become more sensitive to w_0 than any other model parameter [69], so uncertainty in the size of the focused laser will cause error in the best-fit value for k_r. As noted in the Sample Preparation section, it is desirable to minimize the in-plane conductance of the transducer, $k_t h_t$.

[H2] Beam-Offset Measurements of In-Plane Thermal Conductivity:

Beam-offset TDTR is a refinement of the conventional TDTR technique, developed by Feser and Cahill [69,70] that exploits a deliberate lateral displacement between the pump and probe beams to interrogate in-plane heat spreading. In a beam-offset measurement, the signals are measured as a function of the beam offset distance. Beam offset measurement of InSe is shown in Fig. 4(f) [96]. The FWHM of the out-of-phase signal vs. offset distance provides a direct measure of the thermal penetration depth in the in-plane direction [70], $d_{p,r}$.

A key feature of beam-offset measurements is that the measured FWHM along a specific direction is nearly independent of the material's k in orthogonal directions. For example, consider beam-offset measurements of (110) quartz performed by Feser and Cahill [70]. The FWHM of the out-of-phase signal along the c-axis is ≈20× as sensitive to the k of quartz along the c-axis than along the a-axis. Thus, this technique provides an excellent way to measure anisotropic k tensors, such as in quartz [70] , ZnO [105] or black phosphorus [106,107].

The sensitivity of beam-offset measurements to k_r is maximized when $d_{p,r}$ is equal to w_0. In the limit $d_{p,r} << w_0$, the FWHM of the out-of-phase signal depends only on w_0, and so the measurement is not sensitive to k_r. Beam-offset measurements also do not work in the opposite limit, $d_{p,r} >> w_0$. In this low-frequency limit, the thermal dynamics approach a steady-state limit without an out-of-phase signal, analogous to SSTR experiments [108]. Therefore, in this low frequency limit, the magnitude of the thermal signals depends on k_r, but the FWHM does not [105]. In steady-state, the temperature profile must be independent of the heat-capacity because the energy-storage term in the heat-equation is zero, $C(\partial T/\partial t) = 0$. Therefore, in the low-frequency limit, $d_{p,r}$ stops being a useful parameter for describing the thermal dynamics.

[H2] Heat Capacity Measurements:

TDTR performed at multiple frequencies can determine both k and C of low k thin-films, or bulk samples. For thin films, one measurement uses f_{mod} such that d_p is comparable to or less than the film thickness h_f, making the result sensitive to $(kC)^n$. If $d_p < h_f$ then $n = ½$; if $d_p \geq h_f$ then $n < ½$ [109]. The other measurement sets f_{mod} so that $d_p >> h$. In this limit, sensitivity to the film's heat-capacity is reduced, because most of the heat is being stored in the underlying substrate. This is illustrated in Fig. 4(g). However, because the film has a low k, the film remains a significant source of thermal resistance (h_f/k_f). By analyzing both measurements, k_f and C_f can be

simultaneously extracted. Olson *et al.* [110] used this approach to measure heat capacities of ITO and FTO films. Wang et al. [97] used it to measure C of ultralow-k fullerene derivatives, the results of which are shown in Fig. 4(h). We note the value of C determined in this way can have large uncertainty, as shown in Fig. 4(h).

For measurements of C for bulk materials, one measurement is conducted with $d_p \ll w_0$ to determine thermal effusivity (see Eq. 2.2). The second measurement is performed with $d_p \sim w_0$, making the measurement sensitive primarily to thermal diffusivity (k/C). Li *et al.* [111] used this approach to determine k and C of bulk Si. Similarly, Wei *et al.* [112] used this approach to measure substrates and spatially map heat capacity in diffusion couples.

[H2] Time-Resolved Magneto-Optic Kerr Effect (TR-MOKE) Measurements:

Analysis of TDTR experiments assumes that (1) heat is only absorbed by the transducer, and (2) that signals probe only the transducer's temperature. If the transducer is semi-transparent, one or both of these assumptions can be invalid, so generally $h_t > 50$ nm is necessary. To overcome this requirement, TR-MOKE experiments use a ferromagnetic transducer whose temperature-dependent Kerr rotation serves as a proxy for thermoreflectance [71]. Because changes in polarization arise only from the ferromagnetic transducer (assuming no other magnetic or birefringent layers in the sample), h_t can be a few nm without introducing background signals. Additional details about TR-MOKE transducers are provided in Supplementary Note 3.

TR-MOKE provides superior sensitivity for frequency-dependent and beam-offset measurements of in-plane transport, because it allows $h_t k_t$ to be minimized. In a typical TRMOKE transducer, e.g. a 10 nm thick Co/Pt multilayer [113], $h_t k_t \approx 10^{-7}$ W/K, which is 10^2 lower than for an Al thin-film. [108]

[H2] Laser-flash TDTR:

A drawback of standard TDTR experiments is that it does not provide direct information about depth-dependent temperature profiles. Such information is useful for improving sensitivity to buried resistances [116] understanding non-equilibrium heat transfer by electrons [117], and accurately resolving multiple through-plane transport coefficients [74]. To address this limitation, TDTR experiments can be conducted in a laser-flash geometry. In a laser-flash TDTR experiment, instead of a single metal transducer serving the dual purpose of absorbing the pump laser (heater-layer)

and serving as an optical thermometer (thermometer-layer), these functions are separated into different layers. Laser-flash TDTR has been used to study transport in metal multilayers [117–119], the thermal resistance of oxide tunnel junctions [84], and the nature of transport in amorphous thin-films[74]. The combination of laser-flash TDTR and TR-MOKE enabled the first quantitative measurements of the spin-dependent Seebeck effect [120] in ferromagnetic metals.

Initial laser-flash TDTR experiments gathered extra information about transport by analyzing the time-domain response[133]. This limited the utility of the technique to multilayers thin enough for heat to diffuse across the structure within a few nanoseconds. Recently, Peng and Wilson derived a frequency-domain thermal model that allows for accurate prediction of the accumulated response in a laser flash geometry [73], which enables thicker samples to be studied as well[74].

In a laser-flash TDTR experiment, the in-phase signal measures the time-of-flight of heat across the sample, similar to the thermal response from a single pulse. An example of TDTR measurements in a laser-flash geometry are shown in Fig. 5(c-d). Data is shown for a-Si layers of different thicknesses and k = 0.6 W m^{-1} K^{-1}. Analogous to beam-offset measurements of $d_{p,r}$, laser-flash TDTR measurements provide a direct measure of $d_{p,z}$. This is because the phase and amplitude of the temperature response from harmonic heating is governed by z/d_p.

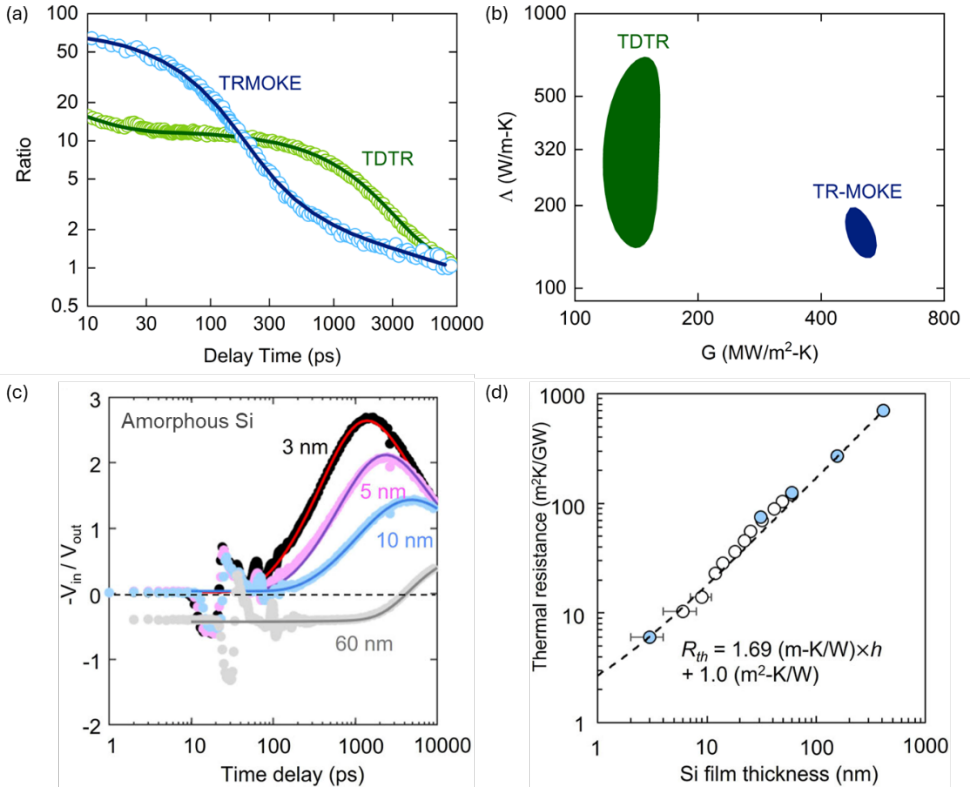


Figure 5: Representative data for TR-MOKE and Laser-flash TDTR. (a) Thermal conductivity measurements of a 0.4 µm AlN thin film using different methods: TR-MOKE with a [Pt/Co]/TiN transducer (blue circles) and TDTR with an 80 nm Al transducer (green circles). Solid lines represent best-fit predictions from the thermal model [72]. (b) Contour plot showing combinations of thermal conductivity and interface conductance that yield fits to the data in (a) with a root-mean-square percentage error below 6.5% [72]. TR-MOKE measurements yields a much lower uncertainty in the AlN thin-film's k. (c) Front/back TDTR data for amorphous Si layers of different thicknesses and k = 0.6 W m^{-1} K^{-1}. The circles are the experimental data and the lines are the best-fit predictions by the thermal model. The ratio shows the time-of-flight of heat across the sample [74]. (d) The thermal resistance of amorphous Si layers as a function of film thickness. Open circles denote measurements a wedge sample, while filled markers denote measurements of samples with a homogeneous film thickness. The observed linear relationship between the thermal resistance and film thickness suggests a thickness-independent thermal conductivity for amorphous Si of 0.6 W m^{-1} K^{-1} [74].

[H1] Applications

In this section, we highlight studies demonstrating TDTR's versatility in measuring thermal properties across diverse systems.

[H2] Thermal interface conductance:

Interfaces are critical to nanoscale thermal transport and high thermal interface conductance, G, is desirable for optimizing thermal management in modern electronic systems [121–125]. TDTR has refined our understanding of G [13,14,66,126–128], by making it possible to test theoretical models for G [129–132]. This is because TDTR probes characteristic length-scales comparable to the Kapitza lengths of most interfaces, $L_K = k/G$. The Kapitza length is the thickness of material that would produce the same temperature drop as the interface does.

TDTR studies have established that most metal/solid interfaces fall in a narrow range between 40 and 400 MW m^{-2} K^{-1} [14,127,133,134]. However, a number of systems are observed to have low conductance (G < 40 MW m^{-2} K^{-1}) due to weak bonding and/or phonon focusing **[G]** [22,48,51,52,134–137]. Alternatively, a number of strongly bonded materials with high speeds of sound have high conductance (G > 400 MW m^{-2} K^{-1}) [94,118,124,138].

Measuring the temperature and/or pressure dependence of material properties is essential for understanding their microscopic origins. As an optical, non-contact method, TDTR can be directly applied to samples in diamond anvil cells (DAC), cryostats or high-temperature stages [94,139–144]. For example, pressure- [142] and temperature-dependent [94] TDTR measurements of G have played a key role in testing theory predictions about the effect of inelastic processes on interfacial heat-currents [145–156].

[H2] Thermal Conductivity:

[H3] High Thermal Conductivity Materials. Understanding thermal transport mechanisms in high k semiconductors (GaN, AlN, diamond, BN, BAs) is essential for high-performance electronic and photonic applications [114,115]. TDTR's micron-scale resolution makes it well-suited for characterizing boron-based crystals, for which it is an ongoing challenge to grow large crystals that are homogenous and have high purity [157]. Recent studies have used TDTR to probe isotropic high k in boron-based crystals, including cubic BAs [40,41,158–160], natural and isotope-enriched cubic

BN [161], and BP [162,163]. Notably, TDTR enabled the thermal characterization of cubic BAs [164], a material predicted by *ab initio* theory to have a k exceeding 1000 W m^{-1} K^{-1} [165–167]. Hou *et al.* report the k of isotopically enriched BAs to be ≈1500 W m^{-1} K^{-1} [159]. The ability to perform TDTR on BAs within a DAC enabled the discovery of unusual pressure independence of BAs k due to four-phonon scattering physics [158,160,168].

Measuring k in high thermal conductivity semiconductor thin films is essential, as their conductivities are often reduced from bulk values due to phonon scattering from size effects, doping, defects, and boundaries [169–171]. The sub-micrometer resolution of TDTR enables measurement of thin-film k[171,172]. However, isolating the high k of semiconductor thin-films from interface conductances is challenging. Keeping heat confined to the film is also challenging. One approach to overcome these challenges is by measuring the thermal conductivity of suspended thin-films [173]. Another option is to prepare films on low-thermal conductivity substrates, then choosing f_{mod} and w_0 so that heat is transported in the in-plane direction [174,175]. Jiang *et al.*[174] used a small laser spot size and low modulation frequency to confine heat within 1–10 μm thick Si thin films on SiO_2 substrates, measuring thermal conductivities between 100 and 150 W m^{-1} K^{-1}. Another method is to improve the measurement's depth-resolution. For example, the ~5-10 nm ferromagnetic transducers used in TR-MOKE measurements confine heat close to the sample surface at short delay times, as discussed in the Results section.[72]

[H3] Ultra-low Thermal Conductivity Materials. Prior to the development of TDTR, it was believed that the minimum k of a fully dense solid was governed by the amorphous limit, as described by the Cahill-Pohl model [176]. The Cahill-Pohl model assumes that, in a disordered solid, vibrational modes will have mean-free-paths equal to half their wavelength. Shortly after its development, TDTR disrupted this old paradigm by identifying a number of thin-film materials whose k is dramatically lower than this theoretical limit [97,177,178]. For example, Chiritescu *et al.* [179] used TDTR to measure ultra-low cross-plane k in layered WSe_2, (~0.05 W m^{-1}K^{-1}), six times lower than the predicted minimum. Subsequent theories suggested that a high degree of elastic anisotropy plays a critical role in suppressing the k in the cross-plane direction [144]. This was tested experimentally through TDTR measurements on muscovite mica by tuning its elastic anisotropy under high pressures in a DAC [144]. Recent studies have demonstrated that ultralow k in layered systems such as $MoSe_2$ [180], r-MoS_2 [181], and r-WS_2 [181] is closely linked to a reduction in shear

modulus caused by turbostratic disorder [182]. Ultralow cross-plane k has also been observed using TDTR in lattice-mismatched $SnSe_2(MoSe_2)_{1.32}$ heterostructures [183] and ordered intergrowths of SnSe and $MoSe_2$ isomers [184], where incommensurate stacking and structural sequencing suppress phonon transport.

[H2] Thermal Anisotropy:

2D materials are of interest both for probing non-classical heat transport at the nanoscale [185] and for applications in optoelectronics, biosensing, and energy storage [186–188]. Resolving the thermal conductivity tensor in anisotropic materials is challenging, as it requires experimental control of the temperature-fields, and the ability to spatially resolve those temperature-fields[104]. TDTR and its variants have been instrumental in resolving anisotropic k in 2D materials[69,104–106,189,190].

TDTR has several advantages for resolving anisotropy over other techniques. Unlike suspended micro-bridge and 3ω techniques, TDTR avoids complex nanofabrication. TDTR can be conducted on small flakes with dimensions on the order of ~20 μm [104,185]. In comparison to Raman thermometry experiments, which are done under steady-state conditions, the ability to tune f_{mod} and w_0 make it easier to control temperature-fields in TDTR. Raman thermometry measurements typically use the same beam to heat, and to measure temperature. This precludes spatial mapping of the temperature-fields along different crystal axis. Raman measurements also require modelling of the absorption vs. depth, which can introduce uncertainty [185].

Zhu *et al.* and Jang *et al.* used beam-offset TDTR and TR-MOKE to measure the thermal conductivity tensor along the zigzag, armchair, and through-plane directions of black phosphorus (Fig. 6(a)) [106,107], illustrating how structural asymmetry affects thermal anisotropy. Figure 6(b) shows a 2D of the out-of-phase TR-MOKE signal, revealing in-plane anisotropy between the zigzag and armchair axes [106]. Fit to the full width at half maximum of the contours yielded an in-plane $k_{zigzag}/k_{armchair}$ anisotropy ratio of ~3 (Fig. 6(c)) [106]. The through-plane conductivity was measured independently using large-spot TR-MOKE, corresponding to a $k_{zigzag}/k_{through\text{-}plane}$ anisotropy ratio of ~20 [106]. The anisotropy was attributed to structural asymmetry of the puckered orthorhombic lattice, which causes direction-dependent elastic constants and phonon scattering rates.

TDTR has been used to study the in-plane and through-plane thermal conductivities of layered transition metal dichalcogenides (TMDs), including MoS_2, WS_2, $MoSe_2$, and WSe_2, revealing thermal anisotropy ratios of up to ~15 [189]. Notably, the through-plane conductivity and Al/TMD interface conductance showed a clear dependence on pump modulation frequency [189]. Layered TMDs have a phonon bandgap that arises from the large mass contrast between the heavy transition metal atom and the lighter chalcogen atoms [191]. The observed dependence of k_z and Al/TMD interface conductance on pump modulation frequency was credited to a nonequilibrium thermal resistance between different phonon groups within the TMDs [189].

TDTR has also been used to study the effects of structural and compositional disorder on the thermal anisotropy of MoS_2 during lithiation[192]. They observed an increase of the thermal anisotropy ratio with lithiation in bulk Li_xMoS_2 crystals, and attributed that to a combination of phonon-focusing effects caused by disorder occurring at different length-scales in the in-plane and through-plane directions [192].

Additional control over temperature-fields can be gained by beam shaping. A number of research groups have used non-circular heater geometries to probe in-plane anisotropic thermal conductivity [105,190,193]. Elliptical-beam TDTR uses an elliptical pump to create a quasi-1D temperature gradient [105]. Rotating the beam orientation allows directional sensitivity to in-plane conductivity [105]. Jiang *et al.* used this method to extract the full thermal conductivity tensor of β-Ga_2O_3 and reported an anisotropy ratio of ~2 between the [010] and [100] directions [193]. In a separate approach, a one-dimensional beam-offset FDTR geometry combines a line-shaped pump with a spatially offset probe to sample the temperature field farther from the heat source than is possible with Gaussian beams [194]. Xu *et al.* used this method to measure in-plane thermal diffusivity in suspended Si, polymers, and HOPG [194].

[H2] Thermal Property Mapping:

Thermal property mapping is crucial for revealing how microstructural features like grain boundaries, defects and crystallographic anisotropy influence local heat flow in materials [17,195]. TDTR has emerged as a transformative tool for spatially resolved thermal property mapping [65,196]. Unlike EBSD and Raman thermometry, TDTR requires minimal sample preparation and allows for rapid, high-throughput imaging [20,197]. The micron-scale spatial resolution of TDTR mapping has allowed for thermal characterization of features such as grain boundaries [195,198], composition

distribution [15,199–201], and interface quality [202,203]. Pioneering work by Huxtable *et al.* [65] used TDTR to map the k of a Nb–Ti–Cr–Si diffusion multiple, revealing anisotropic thermal conductivity in $(Ti,Nb)_3Si$ grains. Thermal conductivity mapping with TDTR helps distinguish between amorphous and crystalline regions in thermally grown oxides used in aerospace coatings based on their phase-dependent k [20,197].

Combining TDTR mapping with electron microscopy is a powerful method for understanding micro-structure / thermal transport relationships. Sood *et al.*[195] correlated thermal conductivity maps of polycrystalline diamond with EBSD imaging of the microstructure in boron-doped polycrystalline diamond (Fig. 6(d)) to quantify the effect of grain-boundaries on transport[208]. Similarly, by correlating SEM backscattered electron (BSE) and EBSD band contrast micrographs with thermal conductivity maps of a stabilization-annealed Si-YbDS environmental barrier coating, as shown in Fig. 6(e), TDTR tracked phase evolution within the coating [18]. The results demonstrated that volatilization of SiO_2 from YbDS at the surface led to the formation of YbMS [18].

[H2] Thermal Properties of Liquids and Biological Samples:

Measuring the thermal properties of liquids is useful for a diverse array of reasons, from device thermal management[204–206], biomedical applications[207–209], to modeling the interiors of exoplanets[210]. The most common way of measuring k of liquids is the hot-wire method, which is based on electrical heating and temperature sensing of a submerged metal wire. TDTR offers advantages over this approach by requiring only a few microliters of fluid, which eliminates convection effects. TDTR also has advantages over transient grating methods because it is insensitive to the liquid's optical properties [45]. Figure 6(f) shows an example of how TDTR can be used to measure the thermal properties of liquids and liquid solid/interfaces [45]. Hsieh *et al.* used TDTR to measure the thermal conductivity of water–methanol mixtures and found that the addition of methanol reduces the thermal conductivity compared to pure water, with values 10% lower than the 0.55 W $m^{-1}K^{-1}$ measured for pure water [210].

Ge *et al.*[209] measured the interface conductance between water and hydrophilic versus hydrophobic surfaces, helping to establish the maximum vapor layer thickness at hydrophobic interfaces[222]. TDTR's has enabled the characterization of local heat transfer coefficients in electronic device hot spots cooled by microchannels [205], jet impingement [204], and two-phase cooling [206], and the thermal

dynamics of water droplets that interact with hot surfaces [211,212]. Mehrvand and Putnam [206] used a high-speed oscilloscope connected to the lock-in amplifier to measure transient changes in the TDTR ratio during the single-bubble ebullition cycle in a microchannel[219]. Their work identified microlayer evaporation as the dominant heat transfer mechanism, offering key insight into phase-change cooling[206].

A related but distinct application of TDTR is measurement of biological samples, because the sample geometry is similar. TDTR offers key advantages over electrical methods for biological applications by using non-contact heating and temperature sensing [207]. This eliminates challenges related to fabrication and integration, making TDTR better suited for complex or delicate sample environments [207]. Xie *et al.* developed a fiber-based thermoreflectance sensor that guides pump and probe beams through an optical fiber, enabling in-situ TDTR measurements on biological samples with challenging geometries [207]. Figure 6(g) presents a schematic of the sensor. This configuration allows accurate thermal measurements of liquids and bio-materials available in small volumes or located in hard-to-reach regions, and has been used to determine the thermal effusivity of diverse bio and agricultural samples, from butter to pork liver [207].

[H2] Thermal Property Measurements with External Stimuli:

Studying thermal transport in materials whose thermal properties change under external stimuli such as electric fields, carrier concentration, chemical intercalation, mechanical strain, or magnetic fields is central to technologies requiring precise heat control at small length scales. These include thermal diodes, transistors, and switches [202,213,214], as well as thin, flexible electronic devices [215] and spin-based systems [216]. Additionally, thermal measurements under high-pressure conditions are essential for geophysical modeling, as pressure-dependent k influences heat flow predictions for planetary interiors [217].

TDTR has proven well suited for doing time-resolved measurements of the electric field effects on different material systems, e.g. electrochemical intercalation [202] and electric-field-driven domain switching [213]. Applying an electric field to suspended PZT membranes drives ferroelastic domain wall motion on nanosecond timescales [213]. Foley *et al.* used TDTR to measure the resulting change in k, observing a reversible 13% increase associated with reduced domain wall density [213]. Sood *et al.* showed that TDTR can be used not only to measure thermal conductance during

electrochemical cycling, but also to spatially map its evolution in real time across a working device through in-situ measurements on ultrathin MoS_2 during lithium intercalation [202].

The capability to apply magnetic fields during TDTR measurements expands its applicability to systems where thermal transport is affected by field-responsive molecular alignment [214] or magneto-structural coupling [216]. One such system is magnetically responsive liquid crystal networks, in which mesogenic units align anisotropically under a magnetic field and affect the direction of heat flow [214]. TDTR performed under magnetic fields at elevated temperatures allowed direct, in-situ measurement of k changes resulting from this reorientation [214].

[H2] Thermal Characterization of Ion-Irradiated Materials:

TDTR has become a key non-contact, non-destructive technique for studying the impact of ion irradiation on the thermal properties of materials, with applications ranging from semiconductor devices to nuclear reactors [218–220]. Ion irradiation produces a subsurface micron-scale damage layer which is inaccessible to direct-contact probes [221]. By varying the thermal penetration depth, TDTR enables non-destructive depth profiling of k within the irradiated region [221].

The thermal properties of semiconductors exposed to ion irradiation are important for cooling electronic devices [220]. TDTR has been used to examine the effect of Ga+ focused ion beam (FIB) processing on silicon, revealing a reduction in k due to the formation of amorphous regions and structural disorder [220].

TDTR has been used to characterize and identify novel materials for nuclear applications [219], and to quantify radiation-induced thermal degradation in materials relevant to nuclear energy systems [218]. Ion irradiation replicates neutron-induced displacement damage [218]. In tetragonal lithium aluminate (γ-$LiAlO_2$), TDTR measurements have shown that irradiation reduces k by up to 90%, reaching values near 1 W $m^{-1}K^{-1}$ at high fluence [218].

Thermal characterization of ion-irradiated materials offers a controlled means to investigate phonon scattering and the mechanisms that govern thermal transport in disordered systems [221–224]. For example, TDTR measurements of k in irradiated silicon showed that reduction in k due to ion irradiation can be reversed by annealing [223]. This indicates that structural disorder from ion-induced damage and boundary scattering reduces k more effectively than impurity scattering from the ions themselves [220,221,223]. In amorphous carbon films, ion irradiation has been shown to anomalously

enhance k with increasing ion fluence and energy [225]. This enhancement results from stiffening, densification, and increased carbon network connectivity [225]. Finally, TDTR has been used to show how ion-induced point defects, near-surface disorder, bond breaking, and surface roughness influence interface thermal conductance [226–229].

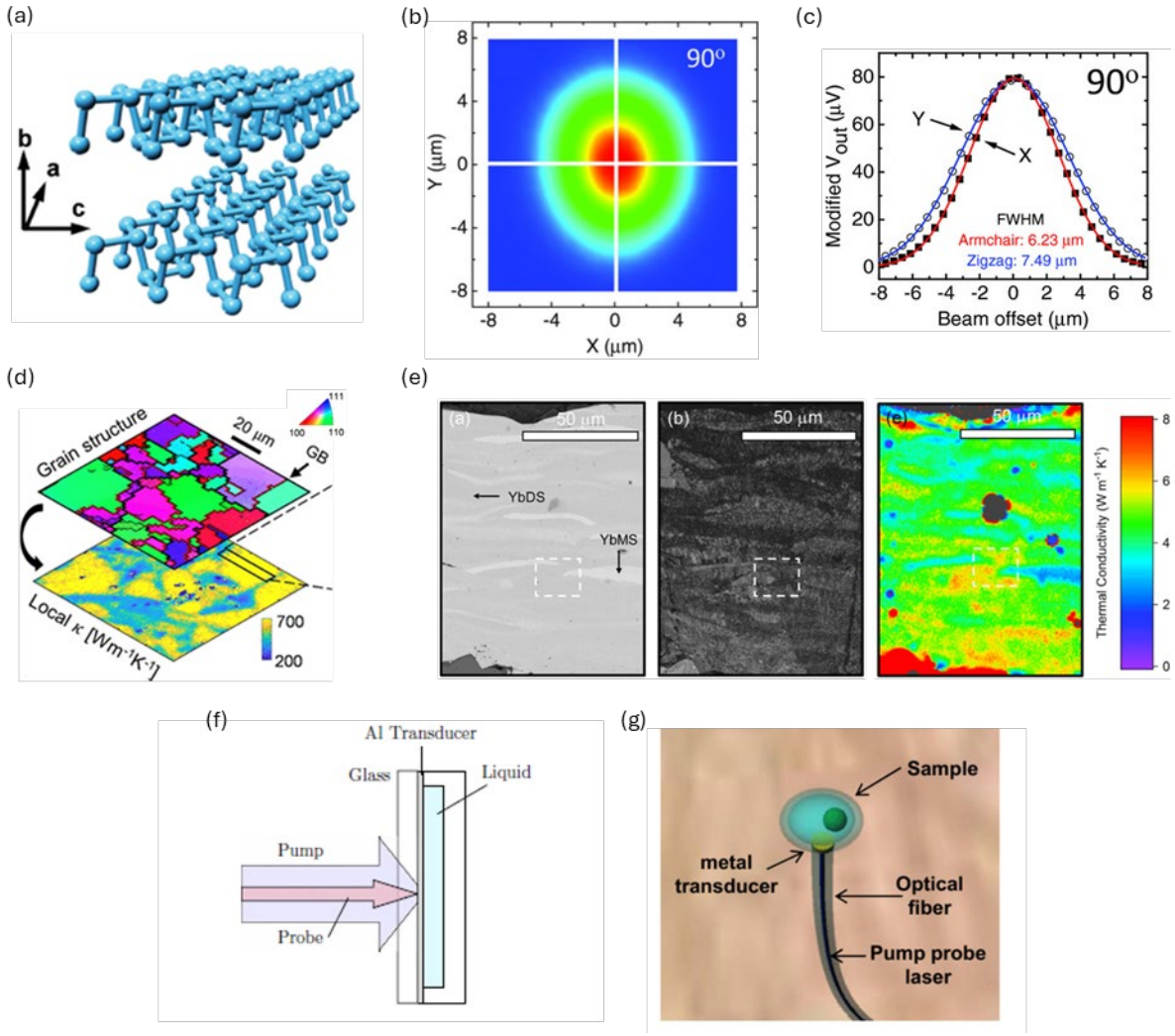


Figure 6: Diverse Applications of TDTR for Thermal Characterization. (a) Schematic of the lattice structure of black phosphorus. The coordinates of *a, b,* and *c* correspond to the in-plane zigzag, through-plane, and in-plane armchair directions, respectively. (b) In-plane beam offset contour measured at 90° of sample-loading orientation with a 20× objective lens and a 1.6 MHz laser modulation frequency to facilitate the in-plane thermal transport [106]. (c) Beam offset signal (open circles and squares) extracted from (a) along the X and Y directions with blue and red lines showing the corresponding thermal model fits. [106]. (d) Correlative EBSD−TDTR microscopy of polycrystalline diamond, where the color-segment indicates the local crystal orientation relative to the x-axis. [195]. (e) SEM BSE and EBSD band contrast micrographs of a stabilization annealed specimen with the corresponding thermal conductivity micrograph with red and yellow regions representative of YbDS and YbMS, respectively [18]. (f)

747 Schematic illustrating how TDTR can be used to measure the thermal properties of liquids [45]. (g) Schematic of fiber-
748 based thermoreflectance sensor that guides pump and probe beams through an optical fiber [207].

[H1] Reproducibility and Data Deposition

TDTR has excellent reproducibility. If the samples are stable, independent measurements on a sample typically differ by less than a few percent. As a result, most TDTR errors are systematic, arising from uncertainties in model parameters, experiment design that does not account for coupled parameters, or spurious signals. In the following sections, we discuss how to manage and quantify these systematic errors, address the importance of accurately determining beam spot-size, identify and mitigate spurious signals, and examine the main noise sources in TDTR.

[H2] Error Analysis:

In TDTR experiments, k can be determined with 5–10% uncertainty, but complexities such as thin films or multiple unknown parameters increase measurement errors significantly. For accurate TDTR experiments, only two unique and independent parameters should be fit per data set. All other model parameters need to be accurately known. To estimate uncertainties, parametric sensitivity analysis or Monte Carlo methods are useful tools. Both of these approaches are expounded further upon in the Supplementary Note 4.

[H2] Spot-size Uncertainty:

Determining the spot size is important for accurately modeling the TDTR signal. There are several ways to determine spot-sizes. The first method involves analyzing the Gaussian intensity of the V_{in} signal using a beam offset scan at high modulation frequencies at a delay time of ~+100 ps. Also, imaging the pump and probe beams with the integrated camera, and then analyzing the resulting beam profile in image analyses software, such as ImageJ will help obtain the spot-size within 5% uncertainty. Lastly, performing a concentric TDTR measurement where $d_p \sim w_0$ on a high k substrate will also yield an independent spot-size measurement. Since high thermal conductivity insulators display mean-free-path effects, high purity single crystal metals work best for this method, e.g. Al or Cu.

[H2] Spurious Signals:

Spurious signals in time-domain thermoreflectance (TDTR) experiments lead to systematic errors by adding unwanted voltage contributions that deviate from the predicted thermal response. Spurious signals should not be confused with noise. Noise can be removed by averaging. Spurious

signals are real signals at f_{mod} that are detected by the lock-in, but that are not related to the sample's thermoreflectance signals. Common sources include coupling between the circuit driving the EOM and the detection circuit, coherent pickup **[G]** by the rf lock-in, partial beam clipping or misalignment in the optical setup, thermoelastic effects in soft materials, and partial transparency of the metal transducer. A number of steps can be taken to avoid spurious signals, such as measuring the specular reflectance, optical chopping, using double-shielded coaxial cables, carefully maintaining beam alignment, and monitoring absolute signals (V_{in}, V_{out}) to diagnose alignment issues. Rough sample surfaces can also cause spurious signals, as explained in Supplementary Note 5.

[H2] Noise:

In TDTR experiments, noise typically stems from three primary sources: electronic noise, intensity noise from the laser, and noise from fluctuations in the amplitude of the EOM modulation. The detection circuit should be designed such that the laser intensity noise dominates over the electronics noise. The laser intensity noise follows a $1/f_{mod}$ dependence at modulation frequencies below 1 MHz [230]. At frequencies near 10 MHz, the intensity noise in a Ti:sapphire laser approaches the shot-noise limit. To mitigate the effects of laser intensity fluctuations at lower frequencies, balanced detection can be used, as shown in Supplementary Fig. S1.

[H1] Limitations and Optimizations

TDTR is a powerful technique, however it has a number of limitations. Due to intensity noise, and pulse accumulation in the in-phase signal (see Supplementary Note 5), TDTR measurements are rarely performed at $f_{mod} < 0.1$ MHz. This has drawbacks, like limiting TDTR's ability to probe buried interfaces. (In Supplementary Note 2, we suggest methods for overcoming this limitation.) In low k samples, e.g. polymers, TDTR's high modulation frequencies ($f_{mod} > 0.1$ MHz) means that it is measuring a small region within 100-200 nm of the surface. This is not ideal for bulk samples where the surface microstructure may or may not be representative of the bulk interior. TDTR does not work for rough samples, which is discussed in greater detail in Supplementary Note 5. TDTR struggles with measurements at very high (> 900 K) and very low (< 30 K) temperatures. At high temperatures, the transducer tends to fail [94,231]. At low temperatures, small heat-capacities make per-pulse temperature rises large compared to the ambient temperature [232,233]. As a result, thermal properties fluctuate with the temperature. The standard analytical approach for analyzing TDTR signals cannot accommodate this. Another drawback of TDTR is that accurate knowledge of heat capacities is required, which can be unknown in novel material systems.

[H2] Mean-Free-Path Effects:

When critical length scales, such as d_p, approach phonon mean-free-paths ℓ, the heat-diffusion-equation becomes inaccurate. As a result, TDTR cannot measure k of non-metallic crystals at low temperatures [234], or with sub-micron w_0 at room-temperature [232]. The heat-diffusion-equation predicts that the heat current scales with $\Delta T/d_p$, where ΔT describes the excess surface temperature of the sample caused by heating. Therefore, the heat-diffusion-equation predicts that ΔT continually decreases with increasing f_{mod} due to ever-increasing temperature gradients. In reality, when d_p approaches ℓ, transport is expected to reach a ballistic limit where the heat-current is proportional to ΔT. Consequently, analyzing TDTR data with the heat-diffusion-equation leads to best-fit values for k (sometimes called "apparent" values) that are low [235,236]. Reducing w_0 to be less than ℓ can induce similar ballistic effects [234]. Frequency-dependent deviations from Fourier's predictions are easiest to observe in alloys with significant point-defect scattering [235]. In contrast, spot-size-dependent deviations are most often observed in single crystals [237].

Over the past 15 years, a number of theoretical studies have clarified this physics. Vermeersch et al. explained trends with f_{mod} in terms of super-diffusive transport [238,239]. Wilson *et al.* emphasized the role of interfaces, suggesting thermal resistance from interfacial scattering extends over length-scales comparable to mean-free-paths [232]. Importantly, Wilson *et al.* argued that this effect is not induced solely by high heating frequency. Instead, high heating frequency increases the measurement's sensitivity to this effect. Recent theoretical studies by Han *et al.* and Li *et al.* support this interpretation [240–242]. Hua et al. also emphasized the importance of interfacial scattering in TDTR measurements of Si [243]. They explained the relatively small frequency-dependent deviations from predictions of the heat-diffusion-equation by considering the dependence of interfacial phonon scattering rates on phonon frequency.

The complex interplay of interfacial scattering and ballistic transport in TDTR experiments offer opportunities for discovery. We believe further progress requires improved spatiotemporal resolution of the temperature-profile, like the kind offered by TR-MOKE and laser-flash experiments. Another important open question is why, unlike TDTR measurements, are SSTR measurements in agreement with diffusive predictions at low temperatures [244], even when mean-free-paths exceed spot size? More work in this space will elucidate the important interplay among different thermometry techniques rooted in impulse, periodic, and/or steady state heating conditions with combinations of in-phase and out-of-phase temperature responses on the ability to resolve intrinsic thermal conductivities of crystals and quasi-ballistic heat transport.

[H2] SSTR for a direct measure of thermal conductivity without the need to know or measure volumetric heat capacity:

Modelling of the thermal response of the sample in a TDTR experiment requires knowledge of the sample's heat capacity. It is sometimes possible to measure heat capacity (as discussed in the Results section), but not always. For example, heat capacity can't be measured if $h_t < d_p$, or in complex heterostructures. Therefore, a direct measure of k is desirable. Recently, Braun *et al.* [108] developed Steady-State Thermoreflectance (SSTR), a pump-probe technique utilizing two continuous wave lasers where the pump beam is modulated at low enough frequencies so the sample reaches steady state conditions during the on-time of the pump. In this approach, the thermoreflectance is monitored as a function of pump-power and compared to a reference standard, which in this steady state regime, becomes a measure of the k of the sample without sensitivity to

the material's heat capacity. Thus, SSTR offers a method to measure k which has been validated on measurements of thermal conductivity on materials such as PCBM [178], with ultralow thermal conductivity, and high thermal conductivity material such as diamond [108], and thermal resistance of thin films and device-relevant heterostructures on length scales overlapping with those of TDTR, providing a complementary path forward for thermal measurements [110,245]. Additionally, SSTR and TDTR used in tandem have recently demonstrated the utility to reduce the uncertainty in determining the thermal properties of thin films and interfaces [172,178,244,246,247]. Thus, in addition to SSTR providing a k measurement without knowledge of heat capacity, these combinations of SSTR with TDTR and their related techniques offer new opportunities for nanoscale thermometry [248].

[H2] FDTR and SSTR for measuring thermal properties of materials deeper under the surface as compared to TDTR:

The thermal penetration depth is usually limited by the relatively high modulation frequency in TDTR [7]. This limiting sample volume thus calls to question the applicability of TDTR to provide a sufficient representation of the thermal properties of materials with characteristic thicknesses that are on the order of several micrometers or greater. While this relatively shallow sampling volume in TDTR should not pose a problem in high quality, homogenous materials (e.g., monolithic wafers), the limitation in TDTR is apparent for material systems with larger length-scale heterogeneous distributions of phases, defects or interfaces where the micron-scale measurement volume of TDTR would not provide a representation of the entire sample. Additionally, the ability to systematically control this depth sampling volume during proper thermal measurements would allow for detecting sub-surface thermal resistances [100,249], including creating thermal conductivity depth profiles, an approach with TDTR measurements that has only recently been demonstrated [221], albeit, still within the relatively shallow sub-surface depth of measurement typical in TDTR. Laser-flash TDTR (see Results section) offers a TDTR variant to measure larger volumes than a typical single sided TDTR measurement, but continuous wave-based pump-probe thermoreflectance alternatives also offer this increased depth sensitivity, discussed below.

The reasonably measurable upper limit of the thermal penetration depth during most-traditionally implemented single-sided pump-probe experiment is on the order of the pump spot size (i.e., the

radius of the "heat source" under the surface) [7,250], which can be accessed in the steady state limit. Thus, SSTR offers an opportunity to thermally probe under a sample surface as compared to typical implementations of TDTR since the thermal gradients induced during an SSTR experiment reach depths on the order of the spot size [251]; this has recently been demonstrated to detect the thermal resistance of buried substrates, interfaces and damaged layers. [247,252,253] The thermal penetration depth could then be varied to sample different volumes by varying the spot size in SSTR, or through implementation of another TDTR-inspired method – Frequency-Domain Thermoreflectance (FDTR).

While frequency domain modulation in continuous wave, laser-based pump-probe methods have been implemented for 4 decades [57,58,254], more recently, Schmidt *et al*. [255] embraced commonalities in hardware, detection and thermal analysis from TDTR that have enabled FDTR to extend to much higher frequencies and broadband modulation [256–261]. By making FDTR measurements over wide ranges of frequencies, the thermal penetration depth is continuously varied up to the upper limit of the spot size, and thus thermal conductivity depth profiles are feasible [262,263]. Additionally, as with SSTR, with larger spot sizes, the FDTR thermal penetration depth can be increased at lower frequencies, thus enabling thermal property measurements of sub-surface films and interfaces at depths deeper than that accessed by TDTR [264–267].

It should be noted that in the low frequency limit, TDTR and FDTR essentially become SSTR, and thus there are opportunities for single thermometry platforms to perform both FDTR and SSTR measurements thus accessing the ability to accurately measure the thermal resistances of multiple films and interfaces with reduced uncertainty as compared to a single technique alone. This tandem approach, coupled with machine learning algorithms, can enable resolution to additional thermal property information with lower uncertainties than previously realized [262,263], offering a unique outlook for AI-ML driven thermometry inspired by TDTR, FDTR and SSTR.

[H1] Outlook

Since the advances in pump-probe experimentation provided by Cahill that have given rise to the now nearly-standardized TDTR configuration detailed in this Primer, TDTR and its offshoots continue to be the most widely utilized thermometry platforms for determining k of thin films and bulk materials along with thermal interface conductance across material interfaces. The continued societal focus and growth of critical technologies that underpin modern and next generation computing and memory technologies, data centers powering AI, energy storage for electric transportation, and communication and radar are all reliant on lower dimensionality materials and devices that have the requisite thermal transport properties to handle the immense power dissipation demands. TDTR will thus continue to be a staple in thermometry to ensure these requisite material thermal transport properties.

As TDTR has inspired evolutions to overcome its various limitations (e.g., SSTR, FDTR, TR-MOKE, beam-offset, and laser-flash TDTR), opportunities exist to continue to advance thermometry through combining the principles of TDTR with other established experimental methods to advance nanoscale thermometry in different directions and towards new extremes. For example, while most common optical spectroscopy platforms interrogate the optical response of materials across ranges of photon wavelengths to understand the electronic, phononic, chemical and physical structure of materials, TDTR interrogates the time and frequency domain thermal response of materials using only a single wavelength or a relatively narrow spectrum of wavelengths (< 10 nm) encapsulated by the spectral bandwidth of the laser pulse (<10 nm). Expanding TDTR to include a broad spectrum of UV, visible and/or IR pump and probe wavelengths will enable its future applicability to wider arrays of materials, including the possibility of using TDTR directly on non-metals without the use of a transducer by exciting electrons above bandgap [87,268] or absorbing directly with optical phonon modes [269–271] (discussed in more detail in Supplementary Note 3). The challenge here, however, lies in harnessing this broad-spectrum photon tuning while still generating an out-of-phase response, which would require higher repetition rate laser systems with sufficient pulse energy for broad-spectrum photon conversion.

As another example, while TDTR has proven exceptional cross-plane spatial resolution in thermal resistance measurements, the lateral resolution is ultimately limited by the diffraction-limited spot

size. Other classes of optical and thermal metrologies achieve true-nanoscale lateral resolution through near-field focusing techniques or scanning probe methods (e.g., Scanning Thermal Microscopy) [272,273], respectively. Certainly, a low barrier to entry for extending the resolution of TDTR to sub-diffraction limit areal resolution could involve patterning transducers with sub-diffraction limit features or using nanoparticles on the surface to achieve this same effect [274–276]. However, an approach more broadly applicable to wider classes of materials would be to combine TDTR with scanning probe methods. There has been immense progress in using optical sources and probes combined with AFM probe tips to interrogate the properties of materials (e.g., TERS [277], SNOM [270,278], nano-FTIR [270], to name a few), but the combination of AFM probes to spatially localize a pump-probe thermoreflectance measurement to length scales beyond the diffraction limit has only recently been explored. This would offer unprecedented nanoscale resolution in TDTR and related techniques. However, as with several of these previous scanning probe-based thermometry techniques, the thermal contact resistance (due to, for example, water and carbon adsorption) and additional near-field radiative heat transfer between the tip and sample surface must be considered for an accurate measure of the nanoscale thermal properties of the sample. Further, localizing the absorption of the ultrafast pump pulse will result in large thermal expansion that can lead to spurious elastic signals, making it difficult to isolate the temperature changes needed to extract the thermal properties of the substrate. Even with these considerations overcome, the thermal diffusion length scales emanating from the AFM-tip heating into the sample must be carefully understood. For example, as discussed above in the Limitations and Optimizations section, the upper limit of the thermal penetration depth in typical pump-probe experiments is on the order of the spot size, so while heating from an AFM tip will certainly increase spatial resolution compared to typical TDTR implementations, tip-surface deformations and anisotropic heating at these nano-length scales could lead to larger heating length scales than anticipated and concomitant decreased spatial resolution. Further, these sub-micron confined heater length scales could also lead to mean free path effects, as discussed in the Limitations and Optimizations section.

Finally, as we discussed in the Applications section of this Primer, the non-contact nature of TDTR lends itself to provide powerful probes of heat transfer in materials being exposed to extreme environments and when external stimuli are applied to materials and basic devices. Most approaches to date rely on designing materials or simple devices that are amenable to the standard TDTR experimental configurations to study these effects. With different optical path hardware and

detection designs, TDTR measurements can be extended to wider arrays of devices and environments to study thermal property changes of materials and devices *in operando* or even in chambers during material growth or material processing. The advent of fiber-optic based pump-probe designs will facilitate this advance [279,280], which could enable, for example real time monitoring of device health in systems, evaluation of defect formation in materials subjected to extreme environments [281,282], or even sensors of material quality during growth that, when coupled with advanced controls and machine-learned algorithms, could provide major advances in high throughput growth of materials with desired optical, electrical and thermal functionalities. Clearly, the future outlook of possibilities of TDTR when coupled with additional hardware and software designs and advances paints an exciting future for optimization of materials, devices and systems with length scales from atom to application.

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Competing interests

The authors declare the following competing interests: Patrick E. Hopkins is co-founder of Laser Thermal Analysis, Inc., a company that has commercialized FDTR and SSTR instruments.

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Glossary:

1. Thermal transport phenomena

Refers to the processes involved in the transfer of energy (i.e., heat) through various media, including solids, liquids, and gases

2. Thermo-reflectance

A phenomenon where the reflectivity of a material changes with temperature, quantified by the material's thermo-reflectance coefficient

3. Thermal model

Refers to the iterative Feldman algorithm modified by Cahill to analyze heat conduction in layered structures using cylindrical coordinates, specifically in scenarios involving surface heating

4. Mean-free-paths

The average distance that a a particle or quasi-particle (e.g., carriers such as electrons, phonons, magnons, etc.) travels between successive collisions that significantly alters its momentum or energy

5. Pulse accumulation

Refers to the systematic collection and averaging of pulsed energy in a material, where the energy from a previous pulse does not fully dissipate prior to the subsequent pulse

6. Picosecond acoustics

Nonthermal signals in pump/probe measurements that arise from strain induced changes to the reflectivity. The initial temperature gradient in the metal transducer after absorption of the pump pulse generates a strain-wave that propagates into the sample. The waves reflect from buried interfaces, and eventually return to the metal film surface. The time-delay of the acoustic waves' echoes (reflections from buried interfaces) is determined by the film thickness and the longitudinal speed of sound.

7. Piezo-reflectance coefficient

A parameter that quantifies the reflectivity response of a material due to applied mechanical stress or strain

8. Multilayer thermal model

Refers to a framework used to analyze the thermal properties of layered materials, by accounting for the interactions between these different layers, by means of quantifying their respective thermal diffusivities and boundary conductances.

9. Phonon focusing

The propagation of phonons preferentially along certain crystallographic orientations in anisotropic materials

10. Thermalize

Refers to the thermal relaxation process in which the longitudinal thermal gradients in a transducer are minimized relative to the thermal gradients across the interface and substrate

11. Coherent pickup

The process of detecting an electrical signal that maintains a consistent phase relationship with its reference signal

12. Acoustic echoes

Refers to the reflections of longitudinal acoustic waves generated by strain pulses that bounce off of interfaces within a material (such as grain boundaries) or between adjacent layers in a multi-layered stack.

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Supplementary information

Time-domain thermoreflectance

In the format provided by the authors and unedited

Supplementary Information for

Time Domain Thermoreflectance

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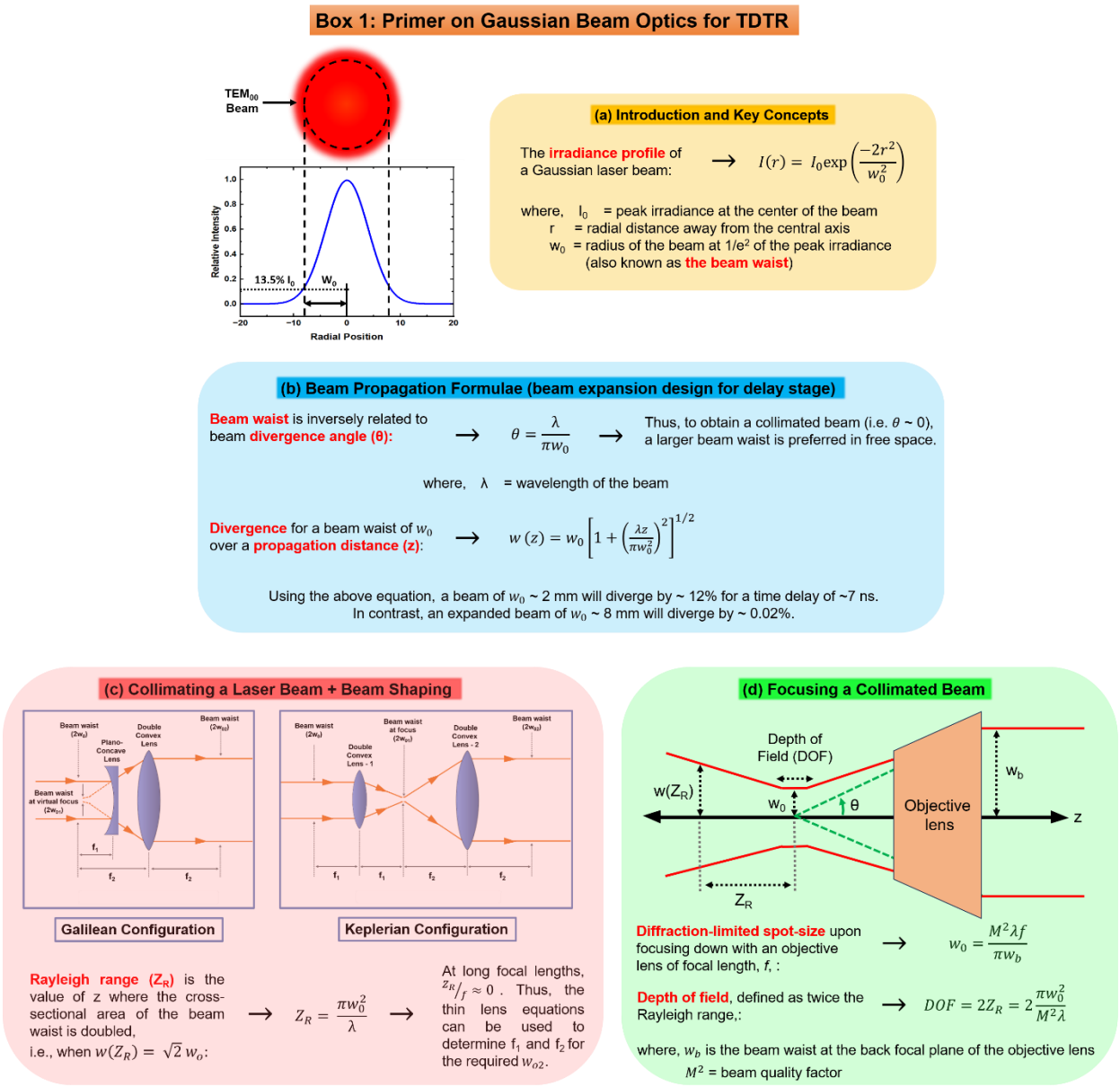
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Supplementary Box 1:



Supplementary Box 1: A Primer on Gaussian Beam Optics for TDTR. We present a short guide to the relevant beam optics when working with beam propagation and beam shaping across a TDTR setup. We first introduce the basic concept of a Gaussian beam; which can be completely described using its beam waist, divergence angle, and irradiance profile. Next, we provide formulae to determine the beam divergence for a given beam waist (i.e., $1/e^2$ spot size), which is crucial to determine how much a beam needs to be expanded before it is passed through a delay stage, to ensure minimum beam divergence and/or wander. We also provide a introduction into the concept of Rayleigh ranges, and demonstrate how to collimate a beam using two standard configurations: the Galilean and Keplerian configurations. Depending on how much table space is available, either of these configurations could be utilized to steer and collimate a beam. Finally, a guide on how to determine the $1/e^2$ spot size at focus is provided (i.e., what is the diffraction-limited spot size possible for a given beam waist in free space). This also aids in determining the M^2 factor of a laser beam off the oscillator.

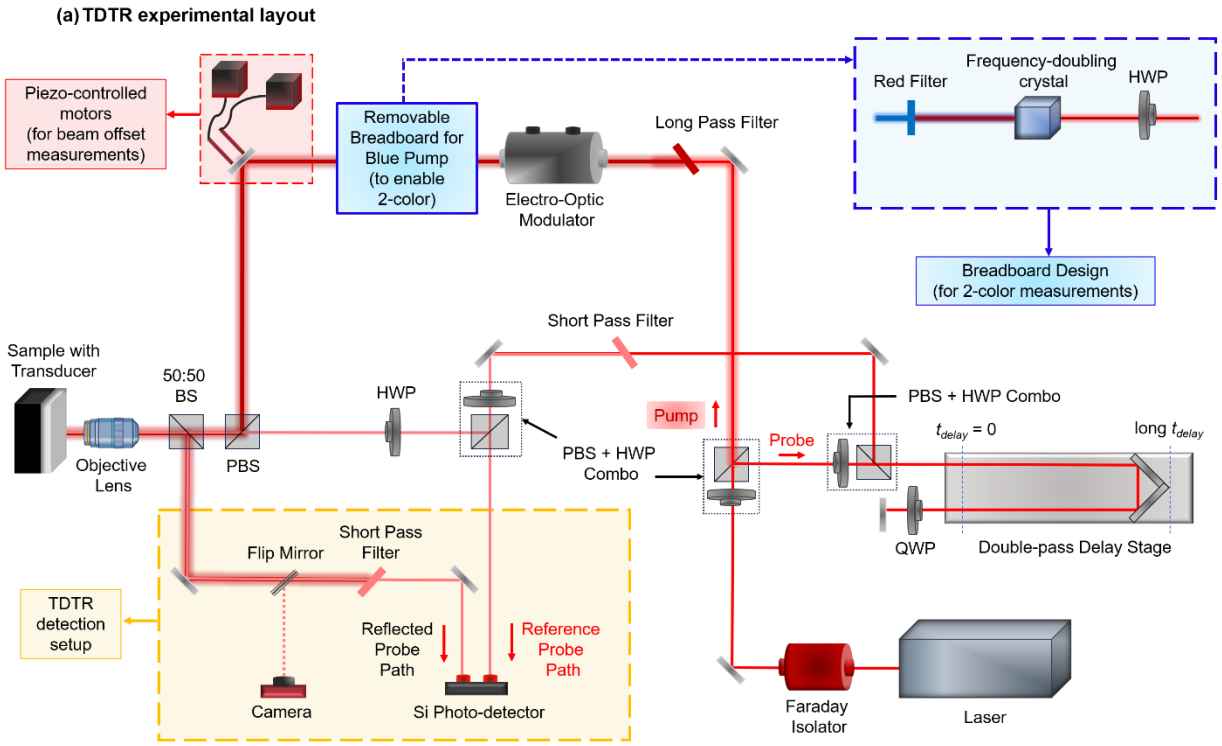


Figure S1. Detailed TDTR layout, with a balanced detection scheme to attenuate laser intensity noise, optical filters for pump beam rejection (two-tint system), or a frequency-doubling system for the pump beam (two-color system).

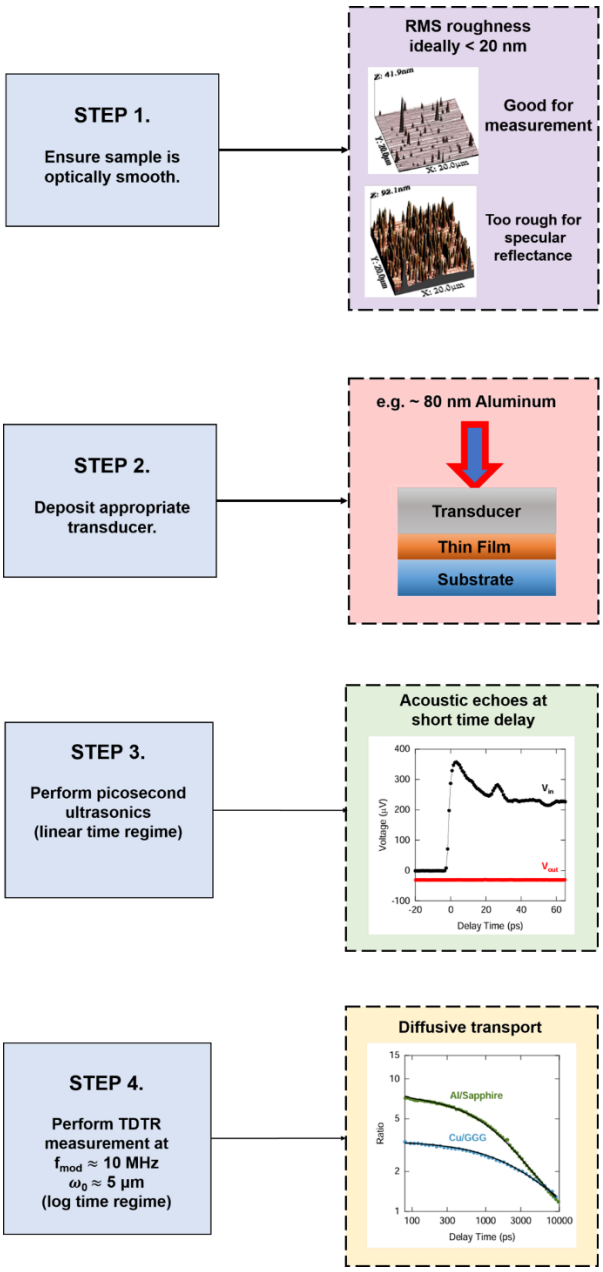


Figure S2. A Four-Step Standard Operating Protocol (SOP) for Standard TDTR Measurements Step 1. involves sample surface characterization to ensure an optically smooth film. We show AFM (atomic force microscopy) images of two sample surfaces – the top sample surface is smooth with a few point defects, suitable for TDTR measurements; the bottom sample surface is rougher and will not yield specularly reflecting beams for a TDTR measurement [1]. A good rule of thumb is to ensure that the rms surface roughness is < 20 nm (or ~λ/40, where λ = pump/probe wavelength). Step 2. involves the deposition of an appropriate transducer – the most commonly used transducer is Aluminum, with a targeted thickness of ~ 80 nm. 4 probe electrical resistivity measurements can be used to characterize the film quality and obtain the thermal conductivity of the transducer. Step 3. involves picosecond ultrasonics measurements to measure the thickness of the deposited transducer, where the acoustic echoes in the phase-corrected Vin signal at short time delay are utilized on an Al/Sapphire sample [2]. Step 4. involves the TDTR measurement at high modulation frequencies (e.g., ~ 10 MHz) and large spot sizes (e.g., 5 μm) to determine k_z and G as a function of pump-probe delay time in Al/sapphire and Cu/GGG (where Cu is the transducer) [2].

Supplementary Table 1:

Technique	Working Principle	Advantages	Limitations/Challenges	Critical Requirements
TDTR	Thermo-reflectance	- Reliable non-contact method to measure k and G across a wide variety of bulk and thin film samples	- Transducer needs to be optically opaque; thus, limiting K and G measurements - Use of high-k thick transducer is severely limiting when measuring in-plane thermal conductivity, especially in thin films - Cannot resolve G when the Kapitza length is less than the transducer thickness (i.e., 80 nm) - Probes only the near surface volume between 100 nm and 10 microns, depending on *k/C*.	- Samples must be specularly smooth (<10 nm). - The transducer must be completely opaque (i.e., the thickness must be 70 – 100 nm)
TR-MOKE	Magneto-optic Kerr Effect	- Use of optically semi-transparent transducer aids in greater sensitivity to both k and G - This technique is particularly useful in k_r measurements of thin films - Could possibly be more forgiving of rough sample	- Difficult to obtain a magnetic transducer with PMA - Determining the thickness of a Co-Pt multilayer transducer stack is challenging - The magnetization of the sample can deteriorate over time.	- Transducer should have low M_s (saturation magnetization) or large perpendicular magnetic anisotropy so it's easy to orient M out-of-plane.

		surfaces on account of +/- field measurements - Less noisy data on account of +/- field measurements	- Measurement with ultrathin transducers < 10 nm are sensitive to *C* of adsorbed molecules from air.	
Beam Offset	Non-concentric beams (can do both TDTR-BO and TR-MOKE BO)	- Usually insensitive to cross-plane conductances in a sample stack; thus, can provide an in-plane k measurement with minimal uncertainty propagation from G and k_z.	- Challenging to measure in-plane k in conductive thin films - Phase jitters at low modulation frequencies lead to significant uncertainties in k_r - Extremely sensitive to spot-size	- Must accurately determine the correlation factor between movement of the mirror/goniometer in radians and the beam offset translation distance on the sample in micrometers.
Laser Flash	Cross-plane separation of heating and temperature sensing.	- Provides a measure of spatiotemporal profile of temperature in cross-plane direction.	- Requires a transducer on both sides of the sample, and sample has to be thin, e.g. 0.1 to 10 microns, depending on *k/C*.	- Requires optically opaque transducers
SSTR	Steady state thermo-reflectance	- Affords greater penetration depths of thermal waves in the sample, thus, allowing for buried interfaces, etc. to be measured. - Does not require requisite knowledge of heat capacity of the sample	- Must accurately characterize the scaling factor correlating the power deposited onto the sample with ΔT.	- Requires optically opaque transducers

Supplementary Note 1: Detector, Detector Circuit and RF lock-in Considerations

Spectral Separation of Pump and Probe Beams. To prevent both specular and diffuse reflections of the pump light from reaching the detector, TDTR setups use spectral separation techniques. This can be achieved using either a two-tint configuration[3] (filters in Fig. S1) where steep-edged filters are employed, or, a two-color configuration where the pump beam is frequency doubled with a BiBO or β-BBO crystal (bread board apparatus in Fig. S1).

A two-tint setup takes advantage of the fact that a ~100 fs laser pulse has a spectral width on the order of 10 nm. Let's assume the center wavelength of the laser is 785 nm, so the laser's power is at wavelengths between 780 and 790nm. A long pass optical filter red-shifts one of the beams (pump beam in Figure S1), by blocking transmission less than some cutoff wavelength, e.g. 787 nm. A short-pass filter blue-shifts the other beam (probe beam in Figure S1) by blocking transmission of power at wavelengths greater than some cutoff wavelength, e.g. 783 nm. Then, the same filter that is on the probe path is placed in front of the detector. In Fig. S1, this is a short pass filter. Since the pump is red-shifted to wavelengths longer than 787 nm, it is blocked, as the short-pass filter blocks power at wavlelengths longer than 783 nm.

A frequency doubled system works similarly. But it uses a single long pass filter on the detector line to block the frequency-doubled pump beam.

By spectrally narrowing the beams, the two-tint configuration broadens the laser pulses to the order of ~1 ps [4].

A down side of spectral separation of the pump and probe beams is there is significant power lost. With the two-tint configuration described in Ref. [4], the pump and probe filters each attenuate the pump and probe power by a factor of 3-4. The attenuation, and filtering efficiency, depends on the filters' cut-off wavelengths. These wavelengths can be tuned slightly by tilting the filters. In power limited experiments, increasing transmitted power by adjusting the filter tilt is beneficial. However, this makes the detector filter less effective at rejecting pump light. Therefore, in these experiments, it is necessary to rely more heavily on spatial and polarization filtering of the pump beam.

For setups employing frequency doubling, power losses are similarly substantial. Typical conversion efficiencies are only 5-10%.

Modulation of the Pump Beam and the RF-lock in. Since achieving the necessary polarization change requires large voltage swings, the EOM is connected to a dedicated driver that amplifies an input signal from a function generator. The driver's bias and driving voltage need to be set to optimize the depth of modulation.

TDTR requires the pump beam induce a thermal response at f_{mod}. To accomplish this, the pump beam can be modulated using either a harmonic or square wave form. The harmonic response at f_{mod} is quantity of interest. However, using a square waveform is acceptable. Square wave heating generates thermal responses at odd harmonics of f_{mod}. Responses at higher order harmonics (3 f_{mod} and above) can by filtered with a digital RF lock-in, e.g., Zurich. Alternatively, an analog lock-in, e.g. the SRS 844, mixes the measured signals with a square-wave at f_{mod}. In this case, unwanted higher-order harmonics must be filtered before lock-in amplification. The higher-order harmonics can be removed with a simple bandpass filter or lowpass filter in the detection circuit.

It is sometimes desirable to measure the thermal response at multiple heating frequencies. Digital lock-ins allow demodulation of multiple frequencies at once.

Since either harmonic and square-wave modulation is suitable for TDTR experiments, the decision about which is best is related to cost and performance of accompanying dedicated driver. For example, the Conoptics 350-160 EOM is commonly used in TDTR systems. It can be driven by an analog driver (model 25A) that can produce a harmonic waveform, or a digital driver (model 25D) that produces a square wave. The digital driver has superior modulation amplitude and bandwidth at a similar cost, therefore many TDTR systems implement a square wave modulation scheme.

Detector. Si photodetectors are operational at wavelengths of ~ 400 – 1000 nm, and feature excellent responsivity at a wavelength of ~800 nm, and thus, are heavily used in TDTR detection set-ups. The reflected probe beam should be focused to a spot much smaller than the detector's active area to avoid clipping, e.g., less than half the detectors area. It should not be focused to such a small area, e.g. tens of microns, that detector nonlinearities are an issue.

Many TDTR setups use a simple Si photodiode. However, balanced detection can reduce noise caused by laser power fluctuations by balancing the signal with a reference beam that does not interact with the sample. This is particularly useful when preforming experiments at low

modulation frequencies, which allows for measurements of thermal diffusivity [5]. When utilizing such a balanced photodetection setup, both the probe beams must be focused down to the same spot-size at the aperture, or detector non-linearities hinder common mode rejection of the noise. The path length of the reference beam and reflected probe beam should also be the same (or at least similar) to optimize common mode rejection.

Step-by-step guide for building TDTR setups. We recommend that the laser beam from the Ti:sapphire oscillator be centered around a wavelength of either ~808 nm or ~785 nm. This is particularly important when designing a 2-tint TDTR setup, because most steep-edged filters are designed to operate around those wavelengths (purchase of optical filters recommended from Semrock, Edmund Optics). Beams at other wavelengths, such as at ~800 nm can be challenging to spectrally separate into low-pass and high-pass configurations, and may require specially designed steep-edged filters, with significant power attenuation.

The laser beam must then be passed through a Faraday Isolator in order to prevent any accidental back-reflections from destabilizing the laser cavity. The laser beam must ideally feature a circular Gaussian profile. If the beams are elliptical, then a pair of cylindrical lenses must be employed to make the beams spherical in free-space. The laser light is then passed through a half-wave-plate (HWP) and polarized beam splitter (PBS) optics pair, where it is separated by polarization into a high-intensity pump and a low-intensity probe beam. The relative intensities of the pump and probe are controlled by rotating the half-wave-plate.

The probe beam path: In our schematic, the probe beam is time-delayed with respect to the pump beam, by passing it through a mechanical delay stage in a two-pass configuration (as shown in the schematic in Fig. 2 in the main document and Fig. S1). A two-pass configuration allows for the correction of any divergence and/or beam wander along the delay stage. In certain other setups, the pump beam is time-advanced with respect to the probe beam, and this allows for lesser artifacts in the signal on account of spot-size changes and/or beam wander as a function of delay time. However, care must be taken to account for the phase-heating function in the measured V_{in} and V_{out} signals, when advancing the pump (details are provided in the Experimentation Section of the main document). Next, the probe beam is sent through a short-pass optical filter, where wavelengths > 785/808 nm are filtered, and only shorter wavelengths are allowed to transmit. This results in a power loss of up to 55%.

The pump beam path: The pump beam is modulated at a desired frequency by passing it through an electro-optic modulator (EOM). Aligning the pump beam through the EOM is absolutely crucial, and will directly impact the quality of data obtained, especially at low modulation frequencies (< 1 MHz). Detailed instructions on how to align the EOM are provided in both the Conoptics resource guide, as well as in online tutorials by Prof. Joseph Feser, and we recommend that the user familiarize themselves with both these resources. To summarize alignment through the EOM, the pump beam must be converged and collimated using a pair of collimating lens such that it is small enough to enter the narrow aperture of the EOM (beam diameter < ~2 mm). Next, a half-wave-plate (HWP) placed before the EOM, will ensure that the polarization of the beam is optimized for maximum modulated transmittance at 50% EOM duty cycle, without having to rotate the EOM itself. Rotating the EOM can sometimes cause a distortion in the beam shape, resulting in an elliptical pump beam. A function generator is used to send a square-waveform to the EOM ($0 \leq f_{mod} \leq 20$ MHz), while certain setups also utilize a sinusoidal waveform to modulate the beam. A well aligned pump beam through the EOM will feature at least ~45% modulated transmission, and will be spherical in free-space. The pump beam is then sent through a long-pass optical filter, where wavelengths < 785/808 nm are filtered, and only longer wavelengths are allowed to transmit. This results in a power loss of up to 65%. We recommend using optical filters with AR-coatings (anti-reflective coatings), to prevent stray back-reflected beams from causing signal artifacts in the final data.

Pump/Probe beams at the sample stage: The pump and probe beams must be well-collimated before they are focused down onto the sample surface using infinity-corrected long-working distance objective lens. A portion of the reflected pump/probe beams can be directed onto a camera to image the surface of the sample (an extensive summary of beam-sample interactions is provided in the Experimentation section of the main document).

Reflected probe beam at the detector: The pump and probe beams are both reflected from the sample surface, however, a TDTR experiment only utilizes thermoreflectance signals from the reflected probe beam. The pump beam must therefore, be completely rejected, through means of either a band-pass filter (in a 2-tint setup) or a blue filter (in case, the pump beam is frequency-doubled in a 2-color setup). The probe beam is then focused down to the aperture of a balanced Si photodetector, using a long-distance focal lens. An additional “reference” probe beam, that doesn’t

interact with the sample, is sent through another long-distance focal lens (with the same focal length as that in the reflected probe beam path) to the other aperture of the photodetector. The resulting RF (radio frequency) voltage signal sent to the RF-lock-in amplifier will feature enhanced SNR (signal to noise ratio) on account of the noise suppression because of this balanced detection scheme.

Supplementary Note 2: Signal Theory and Choosing a pump modulation frequency

Signal Theory. To gain a physical intuition for the in-phase and out-of-phase signals in a TDTR experiment, its useful to compare the TDTR thermal response to the response of simpler heating functions. We examine how the temperature response from TDTR's harmonically modulated train of pump pulses compares to simpler heating scenarios. Specifically, we contrast it with the response to a single laser pulse and to purely harmonic heating.

After absorbing a single laser pulse, the metal transducer's initial temperature rise is determined by the volume over which the laser pulse energy is spread, and the transducer's heat-capacity per unit volume C_m. Then, heat-decays into the sample on a time-scale τ:

$$\Delta T_1(t) = \frac{A_0}{\pi w_0^2 h}\frac{1}{C_t} f(-t/\tau) \qquad \text{(S2.1)}$$

where A_0 is the pulse energy absorbed by the transducer, w_0 is the 1/e^2 spot size, and h is the thickness of the transducer. Here, $f(-t/\tau)$ describes the decay of the sample after heating and depends on τ, k, and C.

For high thermal conductivity materials ($k > hG$, where k is the thermal conductivity and G is the interface conductance), the rate at which the transducer's temperature decays is primarily determined by G and heat-capacity per unit area: $\tau = hC_t/G$, and $f(-t/\tau) = \exp(-t/\tau)$. For low thermal conductivity materials ($k < hG$) the decay of the transducer's temperature is limited by the ability of the sample to carry heat away from the transducer/sample interface, and thus, τ is determined by the thermal properties of the substrate.

The thermal response of a 1D semi-infinite solid to harmonic heating at a frequency f, $q(t) = q_f \exp(i2\pi f t)$, is

$$\Delta T_h(t) = \Delta\bar{T}(f)\exp(i2\pi f t), \text{ with } \Delta\bar{T}(f) = \frac{q_f}{2\sqrt{k\pi C f}}(1-i). \qquad \text{(S2.2)}$$

By defining the thermal penetration depth to be $d_p = \sqrt{k/\pi C f}$, it can be seen that Eq. (S2.2) has the familiar form of the thermal version of Ohm's law: ΔT is proportional to the product of the heat-current q_f and thermal resistance d_p/k. We also see that that the temperature response lags the heating function by $\pi/4$ because the amplitude of the in-phase and out-of-phase responses are equal.

The in-phase and out-of-phase temperatures measured in a TDTR experiment caused by heating from a modulated train of pump pulses are constructed from the frequency-domain response of the sample to harmonic heating [1]:

$$\Delta T_{in}(t) = \frac{1}{2}\sum_{m=-\infty}^{\infty}[\Delta\bar{T}(mf_{rep} + f_{mod}) + \Delta\bar{T}(mf_{rep} - f_{mod})]\exp(i2\pi m f_{rep} t), \quad \text{(S2.3)}$$

$$\Delta T_{out}(t) = -\frac{i}{2}\sum_{m=-\infty}^{\infty}[\Delta\bar{T}(mf_{rep} + f_{mod}) - \Delta\bar{T}(mf_{rep} - f_{mod})]\exp(i2\pi m f_{rep} t). \quad \text{(S2.4)}$$

In Fig. S3(a), we show the thermal response of a Cu crystal coated with an Al thin-film transducer to the three types of heating. For this calculation, the laser spot size is set to $w_0 \approx 10$ μm, which is large enough to ensure that heat transfer is one-dimensional in the through-plane direction. This is because the in-plane temperature gradient scales with $1/w_0$, while the through-plane gradient scales with $1/d_p$, so the through-plane heat-current dominates. The interface conductance is set to 135 MW m^{-2} K^{-1}. The pump modulation frequency to $f_{mod} = 1/8\, f_{rep} = 10$ MHz, a choice that minimizes the effects of pulse accumulation in the in-phase signal (discussed more in Supplementary Note 4). Finally, we set $q_f = A_0 f_{rep}/\pi w_0^2$

Figure S3(a) shows that the in-phase signal measured by the RF lock-in a TDTR experiment is, primarily, a measure of the sample's thermal response to a single laser pulse. Alternatively, the out-of-phase signal measured by the RF lock-in is, primarily, a measure of the out-of-phase response to harmonic heating with $f = f_{mod}$. The in-phase signal corresponds to the temperature response synchronized (in-phase) with the pump modulation. It reflects the immediate temperature rise and the energy stored in the sample after each pump pulse. In contrast, the out-of-phase signal lags the pump by 90° and captures the delayed thermal response governed by heat diffusion away from the heated region. The sample acts as a thermal integrator. Energy is deposited in discrete pulses but diffuses away slowly. Over time, the integrated response is similar to that from continuous harmonic heating.

The curves for the single-pulse response vs. TDTR response in Fig. S3(a) deviate at short delay times, since Eq. (S2.1) neglects the time required for the transducer to thermalize **[G]**. At long delay times, Eq. (S2.1) over-estimates the decay rate slightly because it does not include the effect of heat accumulation (temperature rise) at the surface of the substrate. However, the deviation is small, indicating that the time-domain response of the sample has little sensitivity to the Cu's

thermal conductivity. The harmonic response, $Im\left(\Delta\bar{T}(f)\right)$, is also in good agreement with ΔT_{out}, at least at short-delay times.

Mathematically, this result is straightforward to understand by considering the convergence of the series in Eq. (S2.3) and (S2.4) near $t = 0$. For $t \approx 0$, $\exp(i2\pi m f_{rep} t) \approx 1$. Grouping the $+m$ and $-m$ terms in the series, and invoking the identities $\Delta T(f) - \Delta T(-f) = 2i \cdot \mathrm{Im}(\Delta T(f))$ and $\Delta T(f) + \Delta T(-f) = 2 \cdot \mathrm{Re}(\Delta T(f))$, we obtain

$$\Delta T_{in}(t \approx 0) = (Re[\Delta T(f_{mod})]) + \sum_{m=1}^{\infty}\left(\mathrm{Re}\left[\Delta T\left(m f_{rep} + f_{mod}\right)\right] + \mathrm{Re}\left[\Delta T\left(m f_{rep} - f_{mod}\right)\right]\right), \quad \text{(S2.5)}$$

$$\Delta T_{out}(t \approx 0) = -(Im[\Delta T(f_{mod})]) - \sum_{m=1}^{\infty}\left(\mathrm{Im}\left[\Delta T\left(m f_{rep} + f_{mod}\right)\right] - \mathrm{Im}\left[\Delta T\left(m f_{rep} - f_{mod}\right)\right]\right). \quad \text{(S2.6)}$$

Since $\Delta T(f)$ scales as $1/\sqrt{f}$, the $m \geq 1$ terms in T_{in} are individually smaller than the fundamental $\Delta T(f_{mod})$ term yet add constructively, making them collectively significant and accounting for the rapid temporal evolution of the pulsed response. In contrast, the subtractive nature of the series for T_{out} causes high-frequency contributions to nearly cancel. At f_{mod} = 10 MHz, the ΔT_{out} series converges by m=2 with the fundamental term, $Im[\Delta T(f_{mod})]$, comprising roughly 85% of the total. If f_{mod} is decreased, the fundamental term comprises an increasingly greater fraction of T_{out}.

In TDTR experiments, the ratio of the in-phase and out-of-phase response is analyzed. First, we divide Eq. (S2.1) by the imaginary part of Eq. (S2.2). We note that, in a TDTR experiment, $q_f = f_{rep}(A_0/\pi w_0^2)$. Then, we arrive at the following approximation for the ratio signal:

$$-\frac{V_{in}}{V_{out}} = -\frac{T_{in}}{T_{out}} \approx \frac{\Delta T_1(t)}{\mathrm{Im}(\Delta T(f_{mod}))} = \frac{2\sqrt{\pi f_{mod}}}{f_{rep}}\frac{\sqrt{kC}}{hC_t} f(-t/\tau). \quad \text{(S2.7)}$$

This is Eq. 2.2. in the main text. The ratio is proportional to the sample's thermal effusivity, $\sqrt{kC}$, divided by the transducer's heat-capacity per unit area, hC_m. The proportionality constant only depends on the laser repetition rate and modulation frequency, which are precisely known. The dependence on laser powers, sample absorption, and sample thermoreflectance have divided out. Equation S2.7 neglects the effects that G and hC_t have on the out-of-phase signal, which are not negligible. It also neglects the effect of the sample's thermal resistance on the decay of the in-

phase signal, which is significant for low thermal conductivity materials. Therefore, Eq. S2.7 should not be used to analyze data. Data analyses should be performed with the multilayer thermal models described in [6]. Nevertheless, Eq. S2.7 is useful for qualitatively understanding the observed ratio signals.

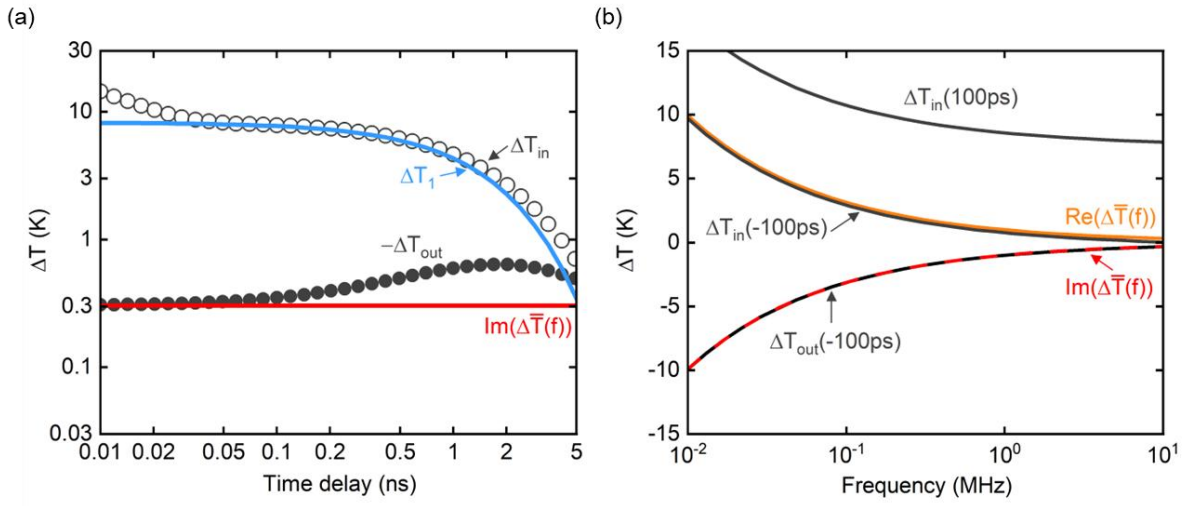


Figure S3: Thermal response in TDTR measurements. (a) Comparison of the in-phase and out-of-phase thermal response in a TDTR experiment, ΔT_{in} and ΔT_{out}, to the expected response for heating by a single pulse, ΔT_1, and harmonic heating, $Im(\Delta\bar{T}(f))$ for a Cu crystal with an Al transducer. (b) Prediction for how the thermal signals in a TDTR experiment, $\Delta T_{in}(-100\ ps)$, $\Delta T_{in}(100\ ps)$, and $\Delta T_{out}(-100\ ps)$, depend on heating frequency f_{mod}. The TDTR signals from pulse accumulation, $\Delta T_{in}(-100\ ps)$ and $\Delta T_{out}(-100\ ps)$, are nearly equal to the response one would expect due to harmonic heating without discrete laser pulses: $\Delta\bar{T}(f)$.

Choosing a Modulation Frequency. In the above discussion of signals, we did not consider the effects of pulse accumulation on the in-phase signal. In Fig. S3(b), the effect of pulse accumulation on ΔT_{in} only appears at delay-times longer than 3 ns, after the pulsed response, $\Delta T_1(t)$, has diminished. The minimal role of pulse accumulation on ΔT_{in} is a consequence of setting $f_{\mathrm{mod}} = 1/8 \cdot f_{rep}$. Fig. S3(c) shows $T_{in}(-100\ ps)$, $T_{in}(100\ ps)$, and $T_{out}(-100\ ps)$. Since ΔT_{out} describes the lagged thermal response of the sample, it is not sensitive to small changes in delay time (i.e., $T_{out}(t = -100\ ps) = T_{out}(t = 100\ ps)$). For simplicity, the calculation in Fig. S3(b) neglects any effects of in-plane transport. Two features in Fig. S3b are notable. First, the change in the in-phase signal: $\Delta T_{in}(100\ ps) - \Delta T_{in}(-100\ ps)$ equals the single pulse response $\Delta T_1(0)$ at all f. So, ΔT_{in} can be viewed as the sum of pulse accumulation (given by $\Delta T_{in}(-100\ ps)$) and the single pulse

response ΔT_1. Second, $T_{in}(-100\ ps)$ crosses through zero at $f_{mod} \approx 1/8 \cdot f_{rep}$. Therefore, choosing $f_{mod} \approx f_{rep}/8$ minimizes pulse accumulation in the in-phase signal.

Minimizing pulse accumulation in ΔT_{in} is advantageous because we want ΔT_{in} to depend primarily on the transducer properties. The reason for this is the following. $\Delta T_{in}(-100\ ps)$which is approximately equal to the real part of the harmonic response at f_{mod}, depends on the sample's thermal properties in the same way as ΔT_{out}. As a result, at low frequencies where $\Delta T_{in}(-100\ ps) > \Delta T_1$, the ratio loses sensitivity to the sample's thermal properties. This is why the default modulation frequency for "standard" TDTR experiments is near 10 MHz, and also (partly) why measurements at low frequencies, e.g., kHz, are uncommon. (The other reason being that laser noise also becomes more problematic at low frequencies.). For many types of measurements, it is advantageous to set the modulation frequency to be much less than $f_{rep}/8$. This adjustment increases the thermal penetration depth, thereby enabling TDTR to determine a variety of thermal properties of samples, such as the in-plane thermal conductivity or heat capacity.

There are two obstacles to low-frequency TDTR measurements. The first is pulse accumulation in the in-phase signal. In some situations, a large value of $\Delta T_{in}(-100\ ps)$ causes the ratio to lose sensitivity to the thermal properties of interest. This can be circumvented by analyzing a modified ratio. The change in ΔT_{in} between -100 and +100 ps will always depend solely on the heat-capacity per unit area of the transducer, regardless of the modulation frequency. Therefore, analyzing $[\Delta T_{in}(t) - \Delta T_{in}(-100\ ps)]/\Delta T_{out}(t)$ instead of $\Delta T_{in}(t)/\Delta T_{out}(t)$ solves this obstacle.

The second obstacle to low-frequency TDTR measurements is noise. Ti:sapphire oscillators become noisy at low-frequencies. The intensity noise can be overcome by using a balanced detection scheme, as illustrated in Fig. S1.

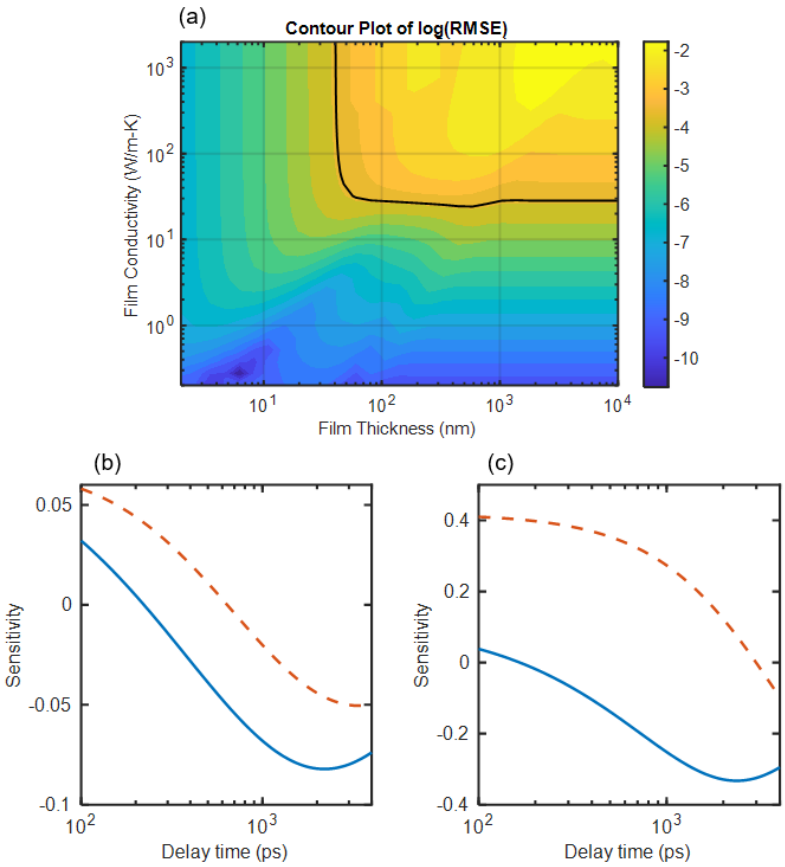


Figure S4: Parameter coupling calculation. These calculations use the TDTR thermal model from Supplementary Ref. 4 to explore whether a TDTR measurement of a thin-film on a silicon substrate can independently determine two parameters: the interface conductance between an 85 nm Al transducer and the thin-film, and thin-film thermal conductivity. The interface conductance was set to 400 MW m^{-2} K^{-1}. (a) Contour map of the natural logarithm of the root-mean-square relative error (RMSE) between the TDTR model and simulated data when the interface conductance is changed to 1.5 × its true value. The bold black contour line marks a 3% error threshold: points below this line (RMSE < 3 %) indicate strong coupling (blue and green), since even with the interface conductance fixed incorrectly, the best-fit thermal conductivity can be adjusted to recover an almost perfect fit (low RMSE). (b) Sensitivity curves for TDTR measurement of a 20 nm film with k = 5 W m^{-1} K^{-1} on silicon. The interface conductance was set to 200 MW m^{-2} K^{-1}. The sensitivity is defined as the fractional change in the ratio due to some fractional change in a thermal model parameter. The solid line is the sensitivity of the ratio vs. delay time to the interface conductance, while the dashed line shows the sensitivity to the film conductivity. The two curves share nearly identical shapes, so changing one parameter can be exactly offset by the other. This indicates the parameters are strongly coupled, and TDTR cannot independently determine them. (c) Sensitivity curves for a 500 nm film with k =30 W m^{-1} K^{-1}. The film conductivity effects the ratio most strongly at short delays, while the interface conductance at long delays. As a result, the two parameters are not coupled, and a TDTR measurement can independently determine both.

Supplementary Note 3: Transducer Considerations and TDTR Without a Transducer

Choosing a Transducer. TDTR experiments usually require coating the sample with a metal transducer. The sample surface before and after transducer deposition needs to be smooth (< 10 nm). The metal transducer in a TDTR experiment serves several purposes. Metals typically have optical penetration depths of 10-20 nm [7]. Therefore, a metal transducer with h_t >50 nm completely confines heating and temperature sensing to the sample's surface. Without a transducer, the laser interacts with the sample over sub-micron or micron scales, depending on the sample's extinction coefficient. A long interaction length makes it hard to resolve nm-scale heat-transfer processes. The transducer also prevents excitation of electron-hole pairs. An opaque transducer also simplifies thermal modeling. Without a transducer, it's necessary to calculate absorption as a function of depth [8]. Additionally, without an opaque transducer, the thermoreflectance signal is a depth-weighted temperature average, not a temperature at one defined location [8,9].

The transducer ensures that heat is evenly spread across the well-defined depth h_t at time delays between ${h_t}^2 C_t/k_t$ and τ. (${h_t}^2 C_t/k_t$ is the time-scale for the transducer to thermalize after laser absorption.) As a result, the only thermal property V_{in} depends on at these time-delays is $h_t C_t$. This feature is fundamental to the operational principle standard TDTR experiments (see Signal Analysis section above). As a result, standard TDTR measurements of the cross-plane k of bulk metals require a transducer (See Supplementary Note X), despite metals having a short optical penetration depth, and no issues with electron-hole pairs.

The following criteria should be considered when selecting the transducer. The figure of merit for maximizing signal-to-noise is $1/R(dR/dT)$ of the metal at the probe wavelength [7,10]. This value typically peaks at wavelengths with interband optical transitions [11]. The transducer resistance $(h/k)$and the transducer/sample boundary resistance $(1/G)$ add to the sample's intrinsic thermal resistances. For cross-plane measurements, h/k and $1/G$ should be small compared to the resistance of interest. For thin-films, the thermal resistance of interest is h_{film}/k_{film}, while for bulk samples its $d_p/k_{substrate}$. The interface conductance G depends on the transducer's bonding strength to the sample, so minimizing $1/G$ means picking a transducer that forms chemical reactions with the sample [12,13]. Minimizing $1/G$ also requires the sample surface be clean prior to transducer deposition, which can be achieved by a RF-sputter etch or pre-deposition bake. Transducers must also be stable. So, its desirable for the transducer to form a self-limiting native

oxide (e.g., Al or Ta), or be inherently inert (e.g., Au or Pt). A large piezo-reflectance coefficient at the probe wavelength is also useful because it enables film-thickness determination by picosecond acoustics. Like dR/dT, piezo-reflectance peaks at interband-transition wavelengths [14].

Aluminum is a good transducer for most experiments with a probe wavelength near 800 nm, where it has large thermoreflectance, strong picosecond acoustic signals, and high k. It also has good adhesion to most samples. For measurements at high T, a refractory metal, e.g. Ta, is a better choice. For high T measurements, it is also a good idea to choose a metal whose thermal expansion coefficient is similar the underlying sample to reduce thermal strain and prevent plastic deformation and delamination.

For TR-MOKE experiments, different factors need to be considered when choosing a transducer. In a TRMOKE experiment, the probe laser reflects from the sample at normal incidence. In this geometry, the probe beam is sensitive to the out-of-plane magnetic moment of the transducer. TRMOKE experiments require the transducer's magnetic moment to point in the out-of-plane direction. If the magnetic moment does not point entirely in the out-of-plane direction, the TR-MOKE signal will not be entirely thermal in origin due to precessional effects [15]. Most magnetic thin films, e.g., Co or Fe, require external fields greater than 1 T to align the moment in the out-of-plane direction. It is experimentally challenging to apply a large out-of-plane magnetic moment, so it is desirable to use a transducer with large perpendicular magnetic anisotropy. This property makes it possible to align the moment in the out-of-plane direction with a weak field that can be supplied with a permanent magnet (< few hundred mT). Ferromagnetic multilayers with alternating sub-nm thick metal layers meet this criterion, e.g., Co/Pt, Co/Pd, CoFe/Pt, or Co/Ni. Rare-earth/transition-metal amorphous alloys, such as GdFeCo or TbFe [16,17] , are another option. Metals with large spin-orbit coupling typically have larger Kerr angles, and therefore better signal to noise ratio in TR-MOKE experiments. Additionally, metals with Curie temperatures that are not to high, e.g. 100-200 K above room temperature, also help with signal-to-noise by increasing dM/dT, and therefore the temperature-dependence of the Kerr angle.

TDTR without a metal transducer. While coating the sample with a thin metal film transducer offers advantages, it can also be a limitation. The transducer introduces an additional thermal boundary resistance at the metal/sample interface, which can reduce measurement sensitivities to the thermal properties of interest, such k of a high thermal conductivity material, k of a sub-surface

thin film, or G of buried interfaces [18]. The transducer can limit the temperature range over which a TDTR measurement can be conducted; most typical metal film transducers used in TDTR will either damage or react with the sample/environment, thus limiting TDTR measurements to an upper bound of around 1000 °C (this upper bound in temperature is strongly dependent on the type of metal film transducer used) [18–21]. To alleviate these potential limitations of using TDTR with a metal film transducer, it is thus promising to conduct TDTR measurements on materials without the metal film transducer, or a non-metal transducer that has higher temperature stability as compared to typical metals.

To this end, transducerless thermoreflectance techniques have been demonstrated for both bulk and thin film semiconductors [22–26]. However, as detailed by Wang *et al*. [22], there are additional experimental and modeling considerations that must be employed. In non-metals, the optical absorption depth is highly dependent on the photon energy relative to the bandgap energy. Non-metals are translucent or transparent to sub-bandgap energy photons, so above bandgap energies for pump and probe should be employed. Above bandgap energies ensure a shorter optical penetration depth. If the photon energy is high enough above the bandgap, the absorption depth can be metal-like, in which case a typical TDTR thermal analysis can be conducted; we note in this case for measurements on semi-infinite materials, only in-plane thermal diffusivity can be resolved with this typical analysis without a transducer [22]. However, without the use of UV sources, likely this scenario will not be the case, and thus the optical penetration depth of the pump and probe must be known and input into the analysis as a depth varying heat source, outlined in prior works.

One of the most important considerations in transducerless TDTR analysis in non-metals is the fact that the reflectivity change is highly dependent on both the temperature and carrier density[27–29]. If the pump changes the carrier density in the conduction or valence bands by exciting long-lived electron-hole pairs, then measured changes in reflectance will depend on both temperature and carrier density. Thermal properties can only be determined from temperature-induced changes in reflectance (thermoreflectance). Thus, carrier modulated changes to reflectance (often termed the “plasma” contribution to reflectivity) must be separated from the measured signals. An advantage of using TDTR to employ transducerless thermoreflectance measurements (as opposed to FDTR, for example), is that the plasma reflectance and thermoreflectance can be separated based

on pump-probe delay time and pump modulation frequency. If using an above bandgap pump, then the plasma component often decays faster than the thermal component, and this difference in decay time becomes more pronounced at lower modulation frequencies. Use of pump energies larger than the bandgap can help mitigate the effect of the plasma contribution to reflectivity by causing a faster decay of excited carriers (these criteria are reviewed by Salnick and Opsal) [27]. Thus, by conducting a transducerless TDTR measurement using above bandgap pump and probe energies at relatively low pump modulation frequencies, the majority of the measured change in reflectivity over the pump-probe delay time will be dominated by the thermoreflectance, as the plasma reflectivity component will be extinguished. Alternatively, the excited carrier contribution to reflectivity can be modeled [30]. This requires including a carrier diffusion equation that is coupled to the heat equation to analyze the data. This introduces additional parameters, and therefore can increase experimental uncertainties. However, if the electron dynamics of the excited non-metal are well known, in addition to the excited state optical properties at the pump and probe wavelengths, then the plasma vs. thermal contributions to the signal can be accurately modeled.

The above discussion focused on understanding the electronic absorption properties to conduct tranducerless TDTR, where the photon energies of the pump and probe should be as much above bandgap as possible to reduce the absorption depth and decrease the excited carrier lifetimes (thus reducing the plasma reflectivity signals); in short, this would typically mean using higher energy visible or UV sources in TDTR. However, recent work has discussed the concept of phonon thermoreflectance [31], where the temperature dependence of the optical phonon modes in polar materials can be used as a thermoreflectance probes. In this case, transducerless TDTR could be conducted by tuning the photon energy of the pump and probe to these infrared active phonon modes. In this case, there is little-to-no free carrier excitation, and the resulting change in reflectance that is measured is a pure thermoreflectance signal (i.e., no plasma component). Recent work has demonstrated the ability to probe these IR-active phonon modes in a pump probe experiment on AlN, SiO_2 and h-BN [31,32]. As with above bandgap experiments, implementing this approach for transducerless TDTR would involve a more specialized laser system as opposed to the standard TDTR Ti:Sapp. Oscillators, specifically with additional OPAs and DFGs to achieve sufficient photon fluences at these mid-IR energies. Further, typical optical penetration depths of radiation at these optical phonon energies are 1-2 orders of magnitude larger than the ~10-20 nm skin depth of a metal [31]. Still, this IR-TDTR approach embracing phonon thermoreflectance offers

the opportunity to conduct transducerless TDTR on materials typically transparent to high energy visible or UV light, such as WBG/UWBG semiconductors and insulators.[89,90]

Supplementary Note 4: Error Analysis

In well-controlled TDTR experiments, thermal conductivity can often be determined with 5–10% uncertainty, but complexities such as thin films or multiple unknown parameters increase measurement errors significantly. In short, there are two requirements for TDTR to be accurate. First, it is necessary to only fit for two parameters per data set, and to confirm that those parameters affect the measured signals in significantly different ways. Second, all other model input parameters need to be accurately known. To quantify how uncertainties in model parameters affect the measured signal, researchers often use either sensitivity analysis or Monte Carlo methods. Sensitivity analysis involves calculating sensitivity parameters, defined as the logarithmic derivative of the ratio. The sensitivity parameter describes how fractional changes in a thermal parameter influence the TDTR signal at different delay times or modulation frequencies. Once calculated with the thermal model, the sensitivity parameters allow for estimates of systematic error propagation. Sensitivity plots also guide experimental design by helping researchers choose laser modulation frequencies and pump–probe geometries that reduce parameter overlap, thereby enhancing confidence in extracted values. Monte Carlo simulations offer another view of how systematic errors propagate in TDTR. In Monte Carlo analysis, researchers assign probability distributions to all relevant inputs, such as sample thickness, heat capacity, and transducer properties, and randomly sample from these distributions to generate many datasets. Each dataset is then fit to the TDTR model, yielding a distribution of best-fit parameters that captures the combined effect of all uncertainties. By examining the spread of this distribution, one obtains confidence intervals.

Supplementary Note 5: Spurious Signals

Spurious signals can cause systematic errors in TDTR experiments, because the accuracy of TDTR depends on matching a thermal model to the measured thermoreflectance signal. Any voltage measured by the lock-in that does not stem from the predicted thermal response is, by definition, a "spurious signal", and will compromise the accuracy of the results.

A major source of these spurious signals arises from coherent pickup in the RF lock-in amplifier. Because the driving signal for the electro-optic modulator (EOM) operates at relatively high voltages (175 V), it is common for there to be weak electromagnetic coupling between the drive circuit and the sensitive detection electronics. Since thermoreflectance signals are on the order of μV, even coupling on the level of 10^{-8} can be problematic.

One approach to reduce the effect of coherent pickup from the driver is to introduce an optical chopper in the pump or probe beam. By modulating the light at a few hundred hertz (i.e., $\pm f_{\text{chopper}}$ away from the EOM frequency), the experimental signal of interest shifts to frequencies that can be more cleanly isolated from the strong driver frequency. However, this approach requires a two-stage lock-in detection scheme. The RF lock-in is operated with a short time constant (to capture frequencies at both f_{mod} and $f_{\text{mod}} \pm f_{\text{chopper}}$). Then, the signals measured by the lock-in are sent to a second audio-frequency lock-in that demodulates at the chopper frequency. The more complicated detection scheme reduces the signal-to-noise ratio by about a factor of two.

Another straightforward step toward mitigating spurious coupling is the use of high-performance double-shielded coaxial cables. We have found that such cables (e.g., those manufactured by Pasternack) are significantly more effective at minimizing pickup than standard coax lines.

Spurious signals can also arise from the optical layout and characteristics of the sample itself. Any clipping of the pump or probe beam can turn small thermoelastic deformations in the sample into unintended intensity modulations at the photodiode, particularly if the beam is partially blocked or deflected. Ensuring a clear optical path by carefully checking the beam footprint as it reflects back from the sample, through the objective lens, to the collection optics, and finally to the detector, will help avoid such artifacts. Similarly, paying attention to the shape of the beams on the camera can alert TDTR users to problems with the beam alignment. Finally, care needs to be taken to

center the reflected beam on the detector, and to make sure the detector area is much larger than the beam size.

Misalignment in the pump–probe overlap or unintentional beam wander as a function of delay time can create errors, especially for measurements of high thermal conductivity materials. Tracking the absolute signals of both V_{in} and V_{out} can be an excellent diagnostic tool for detecting problems. The in-phase voltage is determined by the transducer's optical properties and typically remains similar for a given pump–probe power, regardless of the underlying sample. Unusual deviations in the magnitude of V_{in} may indicate issues with the sample transducer or the optical alignment. Likewise, V_{out} should only vary by about 15–20% over a few nanoseconds of delay, so shifts larger than ~30% often point to alignment problems, such as the delay stage being misaligned or the spot sizes drifting.

A rough surface can be a significant source of spurious signal. Scattered pump light is generally harmless for common two-tint or two-color measurements because it is blocked from reaching the detector by filters. Alternatively, diffuse scattering of the probe can be modulated by thermoelastic effects caused by pump heating [33–38]. Therefore, diffusely scattered probe light will produce non-thermoreflectance intensity fluctuations at f_{mod}. Since this is a thermoelastic effect, it is most problematic in films with large coefficients of thermal expansion. Rough surfaces can also lead to acoustic artifacts that introduce signal via piezo-reflectance of the transducer [39]. Typically, the specular reflectance needs to be at least 95%. In materials with a low coefficient of thermal expansion, e.g., Si or AlN, the specular reflectance can be lower. The specular reflectance is straight forward to measure by monitoring the DC voltage on the detector. Picosecond acoustic signals can serve as another window into surface quality: rough surfaces broaden the acoustic echoes, and significant loss or smearing of these echoes can signal excessive surface roughness and diffuse scattering.

Soft materials, or composites with low elastic moduli, can exhibit additional complexities in short-time data. Lateral stress within the metal transducer can relax via zero-order symmetric Lamb waves, shifting the transducer reflectivity in a manner that falls outside typical thermal diffusion dynamics [40]. These thermoelastic signals are often strongest within the first few hundred picoseconds of the measurement and then die out. When such artifacts are present, analyzing only longer delay data helps to mitigate the error caused by them.

Finally, a key assumption in standard TDTR thermal modeling is that all pump light is absorbed at the transducer surface and that the measured reflectance change is solely a function of temperature in that opaque metal layer. If the transducer is partially transparent or has non-negligible optical penetration by the pump, these assumptions break down. In principle, one can extend the model to include absorption profiles, interference effects, and the thermo-optic response of multiple layers. However, adding these parameters greatly increases the model's complexity and likelihood of parameter correlation. For reliable, straightforward measurements, we find that a metal transducer layer of at least 50 nm thickness is typically sufficient to maintain the standard TDTR assumptions.

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